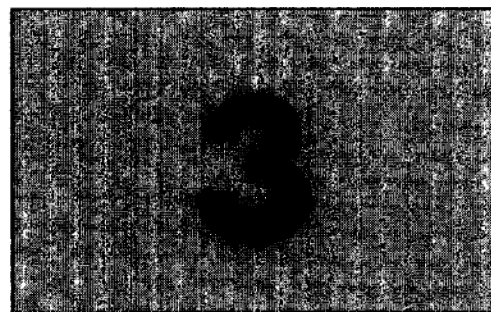


Environments



We have seen in the previous chapter that the magnetic properties of many crystals containing rare earths can be deduced by considering the rare earth ions to be behaving as completely free ions without interacting with each other or their surroundings. However, for magnetic ions in certain crystals one cannot usually ignore such interactions and for many materials they are large and significant. In this chapter we will consider the interactions between an atom and its immediate surroundings. In the following chapter we shall consider the direct magnetic interactions between a magnetic atom in a crystal and its neighbouring magnetic atoms.

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3.1 Crystal fields

To understand the effect of the local environment due to the crystal on the energy levels of an atom, it is necessary to first review the shapes of the atomic orbitals. The angular dependences of the electron density of the s, p and d orbitals are shown in Fig. 3.1. This figure only shows the angular part of the wave functions for each orbital; there is also a radial part (for details, see Appendix C). Only s orbitals are spherically symmetric; the others have a pronounced angular dependence. This is crucial because the local environments are often not spherically symmetric so that different orbitals will behave in different ways.

3.1.1 Origin of crystal fields

The crystal field is an electric field derived from neighbouring atoms in the crystal. In crystal field theory the neighbouring orbitals are modelled as negative point charges; an improvement on this approximation is ligand field theory which is essentially an extension of molecular orbital theory that focusses on the rôle of the d orbitals on the central ion and their overlap with orbitals on surrounding ions (ligands). The size and nature of crystal field effects depend crucially on the symmetry of the local environment. A common case to consider is the octahedral environment. This is because in many transition metal compounds a transition metal ion sits at the centre of an octahedron with an ion such as oxygen on each corner. The crystal field in this case arises mainly from electrostatic repulsion from the negatively charged electrons in the oxygen orbitals. Diagrams of both an octahedral and a tetrahedral environment are shown in Fig. 3.2.

The d orbitals fall into two classes, the t_{2g} orbitals which point *between* the x, y and z axes (these are the d_{xy} , d_{xz} and d_{yz} orbitals) and the e_g orbitals which

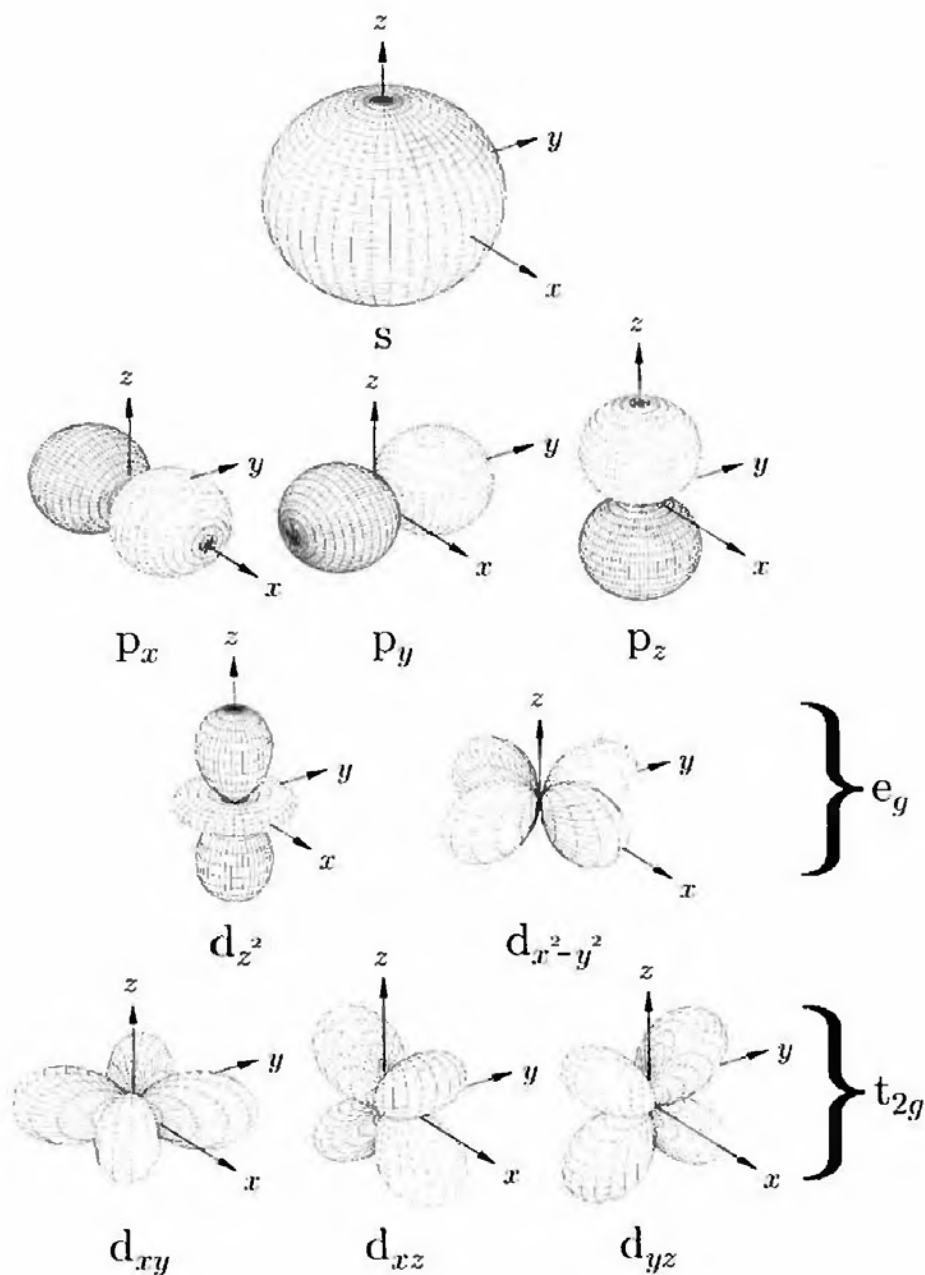


Fig. 3.1 The angular distribution of the s , p and d orbitals. The d_{z^2} and $d_{x^2-y^2}$ levels are grouped together and called the e_g levels. The d_{xy} , d_{xz} and d_{yz} levels are grouped together and called the t_{2g} levels. The d_{z^2} orbital is sometimes referred to as $d_{3z^2-r^2}$.

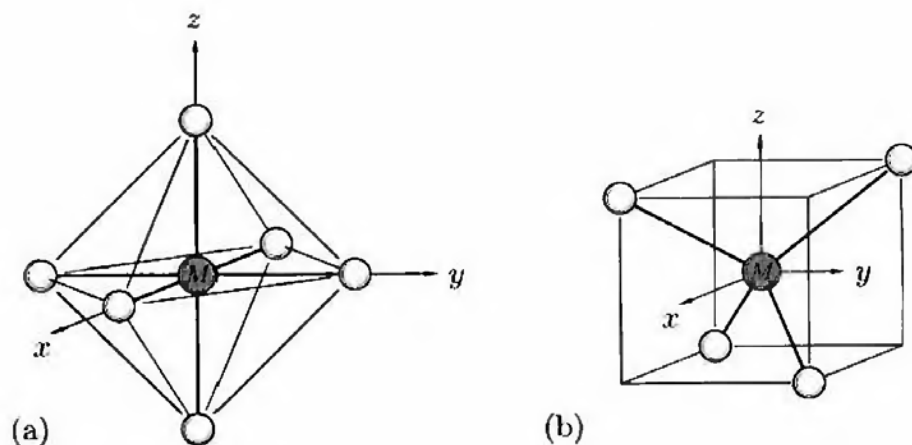


Fig. 3.2 A metal atom M in an (a) octahedral and (b) tetrahedral environment. The octahedral environment is found in many transition metal oxides where oxygen anions sit on the corners of an octahedron with the metal atom at the centre. The tetrahedral environment can be conveniently described by considering alternate corners of a cube as shown.

point *along* these axes (the d_{z^2} orbital, which has lobes which point along the z axis and the $d_{x^2-y^2}$ orbital, which has lobes which point along both the x - and y -axes). Suppose a cation containing ten d electrons is placed in the centre of a sphere of radius r which is uniformly negatively charged. The d orbitals will all be degenerate in this spherically symmetric environment, though the presence of the charge will raise the energy of the entire system. Now imagine the charge on the sphere to collect into six discrete point charges, each lying at the vertex of an octahedron, but still on the surface of the sphere. The total electronic energy of all the d orbitals will not change, but the d orbitals will no longer be degenerate. What has now been created is an octahedral environment.

To demonstrate that the environment affects the orbitals in different ways, consider Fig. 3.3 which shows, in plan view, two different d orbitals in an octahedral environment (this is the projection of Fig. 3.2(a) on to the xy plane). The crystal field is largely produced by p orbitals on neighbouring atoms. It is clear that the d_{xy} orbital (Fig. 3.3(a)) has a lower overlap with these neighbouring p orbitals than the $d_{x^2-y^2}$ orbital (Fig. 3.3(b)) and hence will have a lower electrostatic energy.

In an octahedral environment, neighbouring positive charges congregate at the points $(\pm r, 0, 0)$, $(0, \pm r, 0)$ and $(0, 0, \pm r)$, the three orbitals d_{xy} , d_{yz} , d_{zx} which point between the x , y and z axes will be lowered in energy, but the d_{z^2} and $d_{x^2-y^2}$ which point along the x , y and z axes will be raised in energy. The five levels therefore split as shown in Fig. 3.4(a), with the threefold t_{2g} levels lowered in energy and the twofold e_g levels raised in energy.

If the local environment is something other than octahedrally symmetric, the crystal field may even work in the opposite sense. For example, in a tetrahedral environment, the orbitals which point along the axes now maximally avoid the charge density associated with the atoms situated on four of the corners of the cube which describe a tetrahedron (see Fig. 3.2(b)). Thus in this tetrahedral case the two-fold e_g levels are lower in energy (see Fig. 3.4(b)).

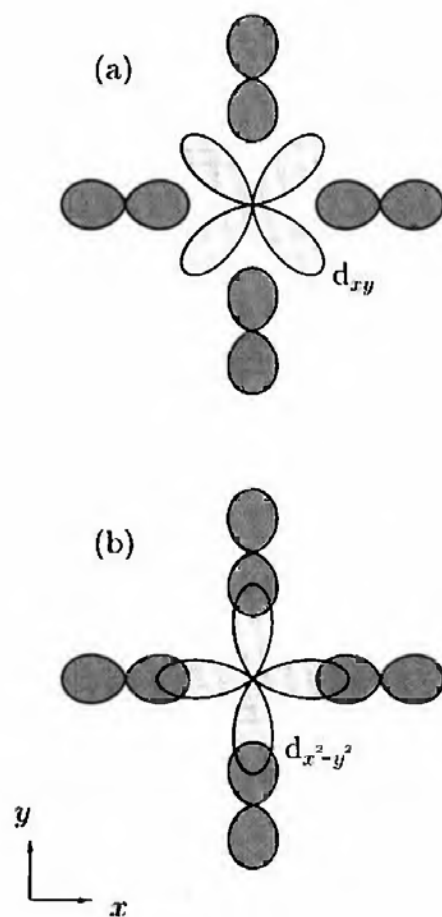


Fig. 3.3 The crystal field originates from an electrostatic interaction. (a) The d_{xy} orbital is lowered in energy with respect to (b) the $d_{x^2-y^2}$ orbital in an octahedral environment.

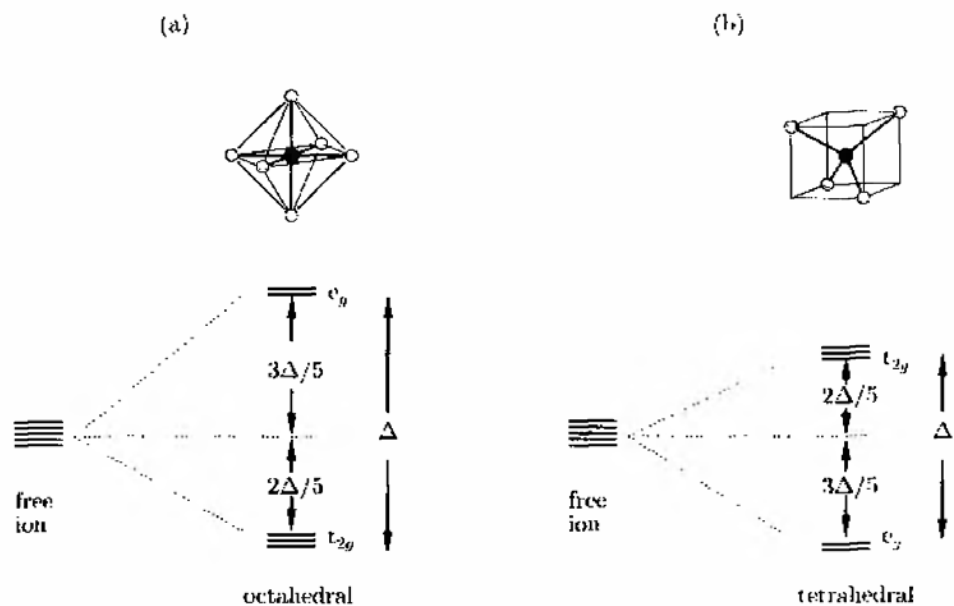


Fig. 3.4 The crystal field in an (a) octahedral and (b) tetrahedral environment.

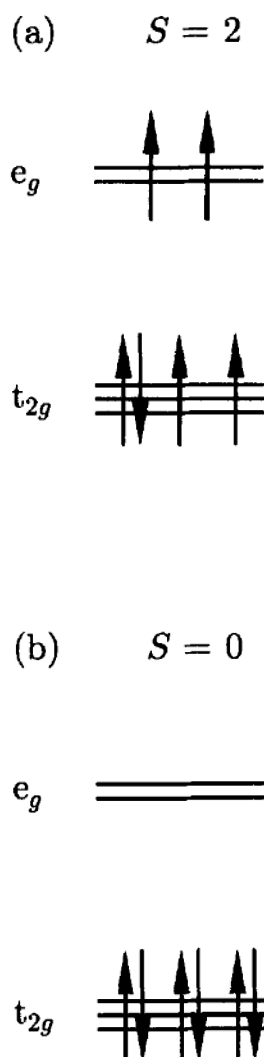


Fig. 3.5 Electronic configurations for the (a) high-spin (weak-field) and (b) low-spin (strong-field) cases for a $3d^6$ ion, e.g. Fe^{2+} .

If one is dealing with a transition metal ion in which not all 3d electrons are present, those that are present will fill the lowest (in this case the t_{2g}) levels before filling the e_g levels. However, the precise order in which the orbitals fill depends on the competition between the crystal field energy and the Coulomb energy cost of putting two electrons in the same orbital, which is known as the **pairing energy** and is usually positive. If the crystal field energy is lower than the pairing energy (known as the **weak-field case**), then as the electrons are added to the system, they will first singly occupy each orbital before any orbital becomes doubly occupied. If on the other hand the crystal field energy is larger than the pairing energy (the **strong-field case**), electrons will doubly occupy the lower energy orbitals before they consider scaling the dizzy energetic heights of the higher energy orbitals. These cases are illustrated in Fig. 3.5.

If one is dealing with octahedral environments, there is never any doubt about how to add the electrons if you are dealing with adding 1, 2, 3, 8, 9 or 10 electrons. The interesting cases occur with 4, 5, 6 or 7. This will now be illustrated for the case of adding six electrons, as would be appropriate for an Fe^{2+} ion (see also Fig. 3.5).

Example 3.1

The Fe^{2+} ion has a $3d^6$ shell. The pairing energy is positive. In the weak-field case, one would fill each orbital once which would leave one electron to spare; this could then be reluctantly paired up with one of the t_{2g} electrons. This would leave four unpaired electrons and a $S = 2$ state (see Fig. 3.5(a)). This is known as a **high-spin configuration**. In the strong field case, all six electrons are shoe-horned into the three t_{2g} orbitals leaving the e_g orbitals unfilled. Since there are no unpaired electrons, the system has $S = 0$ (see Fig. 3.5(b)). This is known as a **low-spin configuration**.

In some materials with Fe^{2+} it is possible to initiate a **spin transition** between the low-spin and high-spin configurations using temperature or pressure or even light irradiation.

3.1.2 Orbital quenching

The expected magnetic ground states for 3d ions are shown in Table 3.1. It is a simple matter to calculate the values of S , L and J by following Hund's rules as outlined in Section 2.5. However, as mentioned there, the values of the predicted moment given by $g_J[J(J+1)]^{1/2}$ do not always appear to agree with experiment. The exception to this is the case of $3d^5$ and $3d^{10}$ for which there is a half or completely full shell of electrons so that $L = 0$.

The reason for this discrepancy is that for 3d ions the crystal field interaction is much stronger than the spin-orbit interaction. Hence Hund's third rule as stated in Section 2.5, which is based upon the fact that spin-orbit interaction is the next most significant energy term after Coulombic effects, is actually wrong. The data appear to suggest that these systems instead choose a ground

Table 3.1 Magnetic ground states for 3d ions using Hund's rules. For each ion, the shell configuration and the predicted values of S , L and J for the ground state are listed. Also shown is the calculated value of $p = \mu_{\text{eff}}/\mu_B$ for each ion using Hund's rules predictions. This is given the symbol $p_1 = g_J[J(J+1)]^{1/2}$ and the next column lists the experimental values p_{exp} which are derived from measurements on paramagnetic salts containing the relevant ions. This agrees much better with $p_2 = 2[S(S+1)]^{1/2}$, which assumes orbital quenching, so that $L = 0$, $J = S$ and $g_J = 2$.

ion	shell	S	L	J	term	p_1	p_{exp}	p_2
Ti ³⁺ , V ⁴⁺	3d ¹	$\frac{1}{2}$	2	$\frac{3}{2}$	² D _{3/2}	1.55	1.70	1.73
V ³⁺	3d ²	1	3	2	³ F ₂	1.63	2.61	2.83
Cr ³⁺ , V ²⁺	3d ³	$\frac{3}{2}$	3	$\frac{3}{2}$	⁴ F _{3/2}	0.77	3.85	3.87
Mn ³⁺ , Cr ²⁺	3d ⁴	2	2	0	⁵ D ₀	0	4.82	4.90
Fe ³⁺ , Mn ²⁺	3d ⁵	$\frac{5}{2}$	0	$\frac{5}{2}$	⁶ S _{5/2}	5.92	5.82	5.92
Fe ²⁺	3d ⁶	2	2	4	⁵ D ₄	6.70	5.36	4.90
Co ²⁺	3d ⁷	$\frac{3}{2}$	3	$\frac{9}{2}$	⁴ F _{9/2}	6.63	4.90	3.87
Ni ²⁺	3d ⁸	1	3	4	³ F ₄	5.59	3.12	2.83
Cu ²⁺	3d ⁹	$\frac{1}{2}$	2	$\frac{5}{2}$	² D _{5/2}	3.55	1.83	1.73
Zn ²⁺	3d ¹⁰	0	0	0	¹ S ₀	0	0	0

state such that $L = 0$ (so that $J = S$, $g_J = 2$) and hence

$$\mu_{\text{eff}} = 2\mu_B \sqrt{S(S+1)}. \quad (3.1)$$

As shown in Table 3.1, this produces a much better degree of agreement with experiment. This effect is known as **orbital quenching** and the orbital moment is said to be **quenched**. For 4f ions, the orbitals are much less extended away from the nucleus and lie beneath the 5s and 5p shells so that the crystal field terms are much less important and Hund's third rule is obeyed. The situation in the higher transition metal ions (the 4d and 5d series) is less clear-cut because the heavier ions have a larger spin-orbit splitting and the effects of the crystal field and the spin-orbit interaction can be comparable.

For the case of the 3d ions where spin-orbit interactions can be effectively ignored, let us consider the orbital quenching effect in more detail. The crystal field in an octahedral environment is given by a constant plus a term proportional to $x^4 + y^4 + z^4 - \frac{3}{5}r^4 + O(r^6/a^6)$ (see Exercise 3.2), and is therefore a real function. This is typical of crystal-field Hamiltonians which can be expressed by real functions (without differential operators) and therefore the eigenfunctions of the Hamiltonian are all real. Now the total angular momentum operator $\hat{L}$ is Hermitian and so has real eigenvalues but the operator itself is purely imaginary.¹ Thus if we have a non-degenerate ground state $|0\rangle$ (non-degenerate so that we can't play tricks by making linear combinations with it and another ground state) which is realized by the crystal-field splitting it must be a real function. Therefore $\langle 0|\hat{L}|0\rangle$ must be purely imaginary because $\hat{L}$ is purely imaginary. But since $\hat{L}$ is Hermitian, $\langle 0|\hat{L}|0\rangle$ must be purely real. It can only be purely real and purely imaginary if

$$\langle 0|\hat{L}|0\rangle = 0 \quad (3.2)$$

¹Equation C.23 shows that the angular momentum operator is $\hat{L} = -i\hbar \times \nabla$ which is purely imaginary.

²This is to make states with azimuthal dependence such as $(\cos m_l \phi)$ or $(\sin m_l \phi)$. Remember that

$$\cos(m_l \phi) = \frac{1}{2}(e^{im_l \phi} + e^{-im_l \phi})$$

$$\sin(m_l \phi) = \frac{1}{2i}(e^{im_l \phi} - e^{-im_l \phi}).$$

so that all components of the orbital angular momentum of a non-degenerate state are quenched.

This result can be seen in another way. Orbital states which are eigenstates of $\hat{L}_z$ with eigenvalue m_l have an azimuthal dependence of the form $e^{im_l \phi}$. The requirement for real eigenfunctions implies that linear combinations of states with $\pm m_l$ must be formed,² so the resulting state has $\langle L_z \rangle = 0$. A semiclassical interpretation of orbital quenching is that the orbital angular momentum precesses in the crystal field, so that its magnitude is unchanged but its components all average to zero.

In fact the orbital angular momentum may not be completely quenched because the spin-orbit interaction is not completely ignorable, even in the 3d ions. The spin-orbit interaction can be included as a perturbation and in this case it can mix in states with non-zero angular momentum. This results in a quenched ground state but with a g-factor which is not quite equal to the spin-only value of 2, the difference from 2 reflecting the mixed-in $L > 0$ states. It can also result in the g-factor being slightly anisotropic so that its value slightly depends on which direction you apply the magnetic field with respect to the crystal axes.

3.1.3 The Jahn-Teller effect

We have assumed so far that all we need to do is to work out what kind of symmetry the local environment has, then deduce the electronic structure

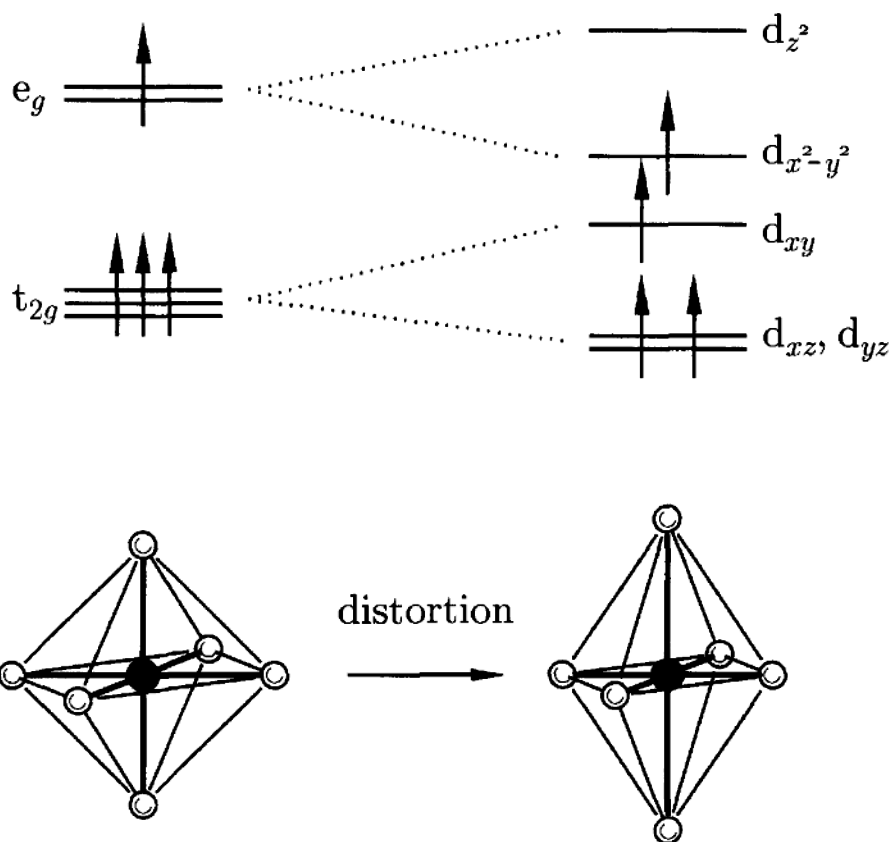


Fig. 3.6 The Jahn-Teller effect for Mn^{3+} ($3d^4$). An octahedral complex (left) can distort (right), thus splitting the t_{2g} and e_g levels. The distortion lowers the energy because the singly occupied e_g level is lowered in energy. The saving in energy from the lowering of the d_{xz} and d_{yz} levels is exactly balanced by the raising of the d_{xy} level.

and hence figure out the magnetic properties based on how many electrons there are to fill up the energy levels. Sometimes, however, the magnetic properties themselves can influence the symmetry of the local environment! This comes about because it can sometimes be energetically favourable for, say, an octahedron to spontaneously distort as shown in Fig. 3.6 because the energy cost of increased elastic energy is balanced by a resultant electronic energy saving due to the distortion. This phenomenon is known as the Jahn–Teller effect. For example, Mn^{3+} ions (which have a configuration $3d^4$) in an octahedral environment show this kind of behaviour (see Fig. 3.6). In contrast, Mn^{4+} ions ($3d^3$) would not show this effect because there is no net lowering of the electronic energy by a distortion.

To describe the effect, at least at the phenomenological level, we will assume that the distortion of the system can be quantified by a parameter Q , which denotes the distance of distortion along an appropriate normal mode coordinate. This gives rise to an energy cost which is quadratic in Q and can be written as

$$E(Q) = \frac{1}{2}M\omega^2 Q^2, \quad (3.3)$$

where M and ω are respectively the mass of the anion and the angular frequency corresponding to the particular normal mode. This relation is plotted in Fig. 3.7(a). Clearly the minimum distortion energy is zero and is obtained when $Q = 0$ (no distortion).

The distortion also raises the energy of certain orbitals while lowering the energy of others. If all orbitals are either completely full or completely empty, this does not matter since the overall energy is simply given by eqn 3.3. However, in the cases of partially filled orbitals this effect can be highly significant since the system can have a net reduction in total energy. The electronic energy dependence on Q could be rather complicated, but one can write it as a Taylor series in Q and provided the distortion is small it is legitimate to keep only the term linear in Q . Let us therefore suppose that the energy of a given orbital has a term either AQ or $-AQ$ corresponding to a raising or a lowering of the electronic energy, where A is a suitable constant, assumed to be positive. Then the total energy $E(Q)$ is given by the sum of the electronic energy and the elastic energy

$$E(Q) = \pm AQ + \frac{1}{2}M\omega^2 Q^2, \quad (3.4)$$

where the two possible choices of the sign of the AQ term give rise to two separate curves which are plotted in Fig. 3.7(b). If we consider only one of them we can find the minimum energy for that orbital using $\partial E/\partial Q = 0$ which yields a value of Q given by

$$Q_0 = \frac{A}{M\omega^2} \quad (3.5)$$

and a minimum energy which is given by $E_{\min} = -A^2/2M\omega^2$ which is less than zero. If only that orbital is occupied, then the system can make a net energy saving by spontaneously distorting.

What we have been considering so far is essentially a static Jahn–Teller effect because the distortion which can spontaneously occur is fixed on a particular axis of an octahedron. However, the distortion can switch from one

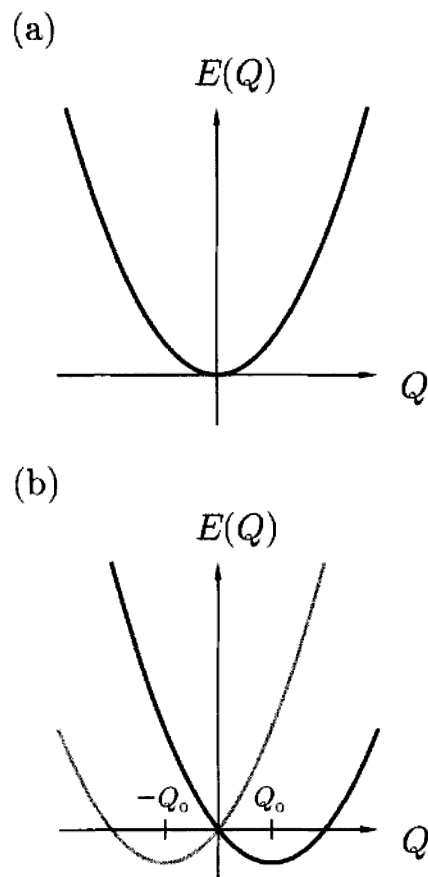


Fig. 3.7 (a) The energy of an octahedral complex as a function of the distortion Q according to eqn 3.3. (b) The energy of an octahedral complex as a function of the distortion Q according to eqn 3.4.

axis to another at higher temperatures, giving rise to a **dynamic Jahn–Teller effect**. Another type of dynamic effect involves rapid hopping of the distortion from site to site. This is important in, for example, materials which contain a mixture of Mn^{3+} and Mn^{4+} ions. Such phenomena can be detected by their effect on magnetic resonance data.

A further effect is that in certain materials (e.g. DyVO_4), below a certain critical temperature, a Jahn–Teller distortion of each complex can occur *cooperatively* throughout the crystal. This is known as a **cooperative Jahn–Teller transition**.

This concludes our discussion of the effect of the crystal field. In the rest of this chapter there follows a description of various experimental techniques which can be used to study the interaction of a magnetic moment with its environment.

3.2 Magnetic resonance techniques

The local environment of a magnetic moment is determined by the crystal fields, but spin–orbit couplings and hyperfine interactions with the nuclei also play an important rôle in controlling the electronic structure. These effects can be studied by using a variety of experimental techniques which involve magnetic resonance. These experimental techniques will be described in the remainder of this chapter.

We have already seen in Section 1.1.2 how the application of a magnetic field to a magnetic moment can induce precession of that magnetic moment at an angular frequency given by $|\gamma B|$ where γ is the gyromagnetic ratio. A system of magnetic moments in a magnetic field can absorb energy at this frequency and thus one may observe a resonant absorption of energy from an electromagnetic wave tuned to the correct frequency. This is **magnetic resonance** and it can take a number of different experimental forms, depending on what type of magnetic moment is resonating. The appropriate frequency of the electromagnetic wave depends on the sizes of both the magnetic field and the magnetic moment. Figure 3.8 summarizes the range of electromagnetic radiation as a function of frequency for a range of different units. It can be used for reference and referred to throughout the following discussion as necessary.

3.2.1 Nuclear magnetic resonance

The most commonly used form of magnetic resonance is **nuclear magnetic resonance (NMR)**. This technique is greatly employed in medical imaging, where it goes by the name of **magnetic resonance imaging (MRI)** to avoid the use of the dreaded word “nuclear”. Some people may be devastated to discover that the average human body contains over 10^{27} nuclei; by mass we are $\sim 99.98\%$ nuclear! MRI measures NMR in the protons in a patient’s body and provides a safe, non-invasive technique of yielding detailed cross-sectional images.

To perform any NMR experiment, one needs a nucleus with a non-zero spin. Nuclei which are commonly studied include ^1H (proton), ^2H (deuteron) and ^{13}C . In a simple NMR experiment (see Fig. 3.9), a sample is placed inside

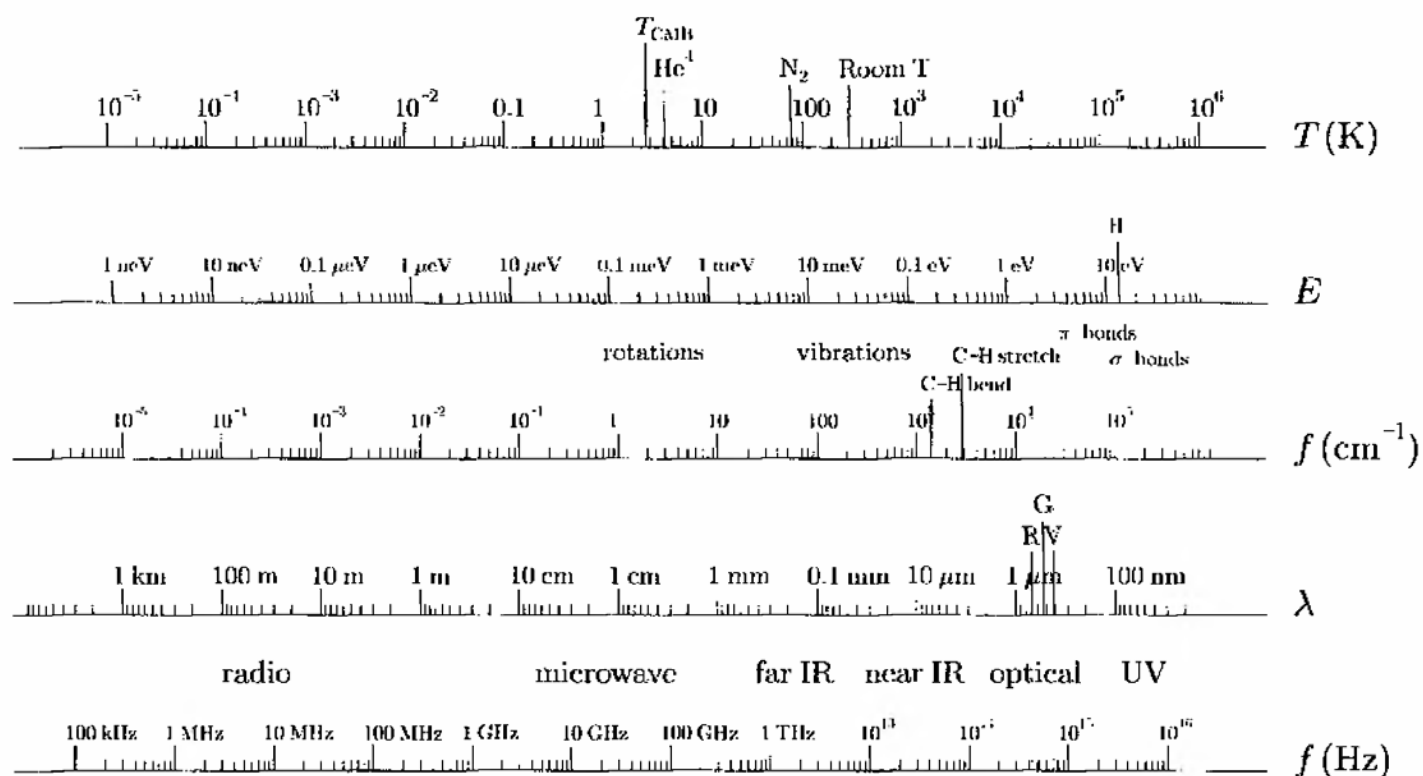


Fig. 3.8 The electromagnetic spectrum. The energy of a photon is shown as a temperature $T = E/k_B$ in Kelvin and as an energy E in eV. The corresponding frequency f is shown in Hz and, because the unit is often quoted in spectroscopy, in cm^{-1} . The cm^{-1} scale is marked with some common molecular transitions and excitations (the typical range for molecular rotations and vibrations are shown, together with the C–H bending and stretching modes). The energy of typical π and σ bonds are also shown. The energy of typical π and σ bonds are also shown. The particular temperatures marked on the temperature scale are T_{CMB} (the temperature of the cosmic microwave background), the boiling points of liquid Helium (He^4) and nitrogen (N_2), both at atmospheric pressure, and also the value of room temperature. Other abbreviations on this diagram are IR = infrared, UV = ultraviolet, R = red, G = green, V = violet. The letter H marks 13.6 eV, the magnitude of the energy of the 1s electron in hydrogen. The frequency axis also contains descriptions of the main regions of the electromagnetic spectrum: radio, microwave, infrared (both ‘near’ and ‘far’), optical and UV.

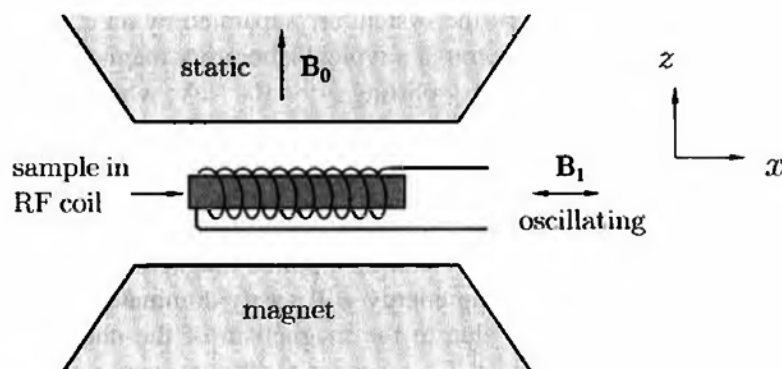


Fig. 3.9 Schematic diagram of an NMR experiment. The sample sits inside a radio frequency (RF) coil which produces an oscillating RF field. A highly homogenous static magnetic field is provided by a magnet. The static field B_0 and the oscillating field B_1 are perpendicular to each other. In a real experiment, the sample would be much smaller than is shown here so that it experiences a uniform field from the RF coil.

a coil which is mounted between the pole pieces of a magnet. The magnet produces a magnetic field B_0 along a particular direction, say the z -direction. We have already seen that the quantity m_I , the z component of the angular momentum of the nucleus, can only take integral values between $-I$ and I .

³The positive charge of the nuclei and the negative charge of the electron is responsible for the differing signs in the expressions:

$$\begin{aligned}\mu &= +g_N\mu_N\mathbf{I} \\ \mu &= -g_J\mu_B\mathbf{J},\end{aligned}$$

for the nuclear and electronic moments respectively.

⁴For the time being, we set $J = 0$ and thus ignore the electron spin.

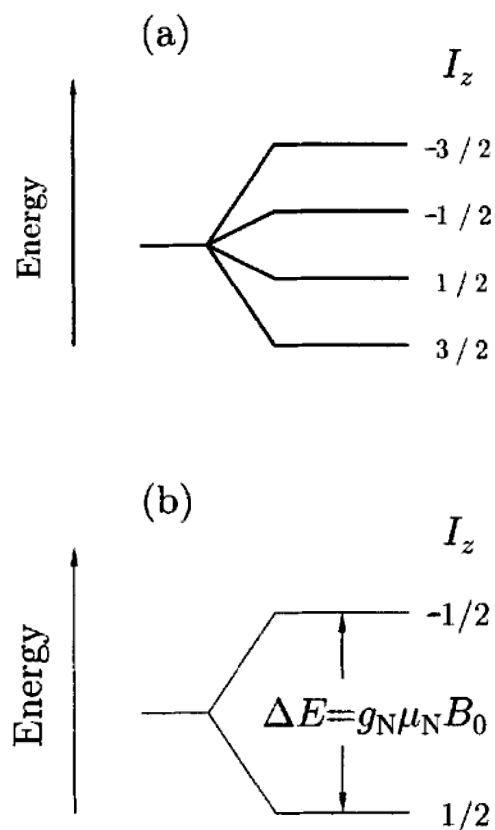


Fig. 3.10 (a) The four levels from an $I = \frac{3}{2}$ nucleus. (b) In the simpler case of an $I = \frac{1}{2}$ nucleus, there are just two levels. These two nuclear levels are separated by an energy $\Delta E = g_N\mu_N B_0$; the lower (upper) level corresponds to the nuclear magnetic moment lying parallel (antiparallel) to the magnetic field B_0 .

⁵Actually you do not have to get it exactly right. You can be plus or minus the linewidth which can be determined by various factors, including variations in the magnetic field at the nucleus due to the dipolar field produced by neighbouring nuclei.

The energy E of the nucleus is the energy of a magnetic moment³ μ in the magnetic field $\mathbf{B}_0$, and thus $E = -\mu \cdot \mathbf{B}_0$ so that

$$E = -g_N\mu_N m_I B_0. \quad (3.6)$$

This corresponds to a ladder of $2I + 1$ equally spaced levels with mutual separation $g_N\mu_N B_0$, as illustrated in Fig. 3.10(a) for the case $I = \frac{3}{2}$, as found in Na or Cu.⁴ Exciting transitions between adjacent pairs of levels with a radiofrequency (RF) field is the basis of nuclear magnetic resonance. The RF field $\mathbf{B}_1$ is applied in the x -direction, and leads to a perturbation of the system which is proportional to $B_1 \hat{I}_x$. The matrix element of the perturbation is proportional to $\langle m'_I | \hat{I}_x | m_I \rangle$ and is zero unless $m'_I = m_I \pm 1$. The allowed transitions are therefore described by the selection rule,

$$\Delta m_I = \pm 1, \quad (3.7)$$

which implies that only transitions between adjacent states may occur. In the most common arrangement, the RF coil not only produces the excitation, but is itself part of a tuned circuit with a large Q factor. As transitions are excited in the nuclei by the RF field, energy is transferred between the RF circuit and the sample, and this results in small changes of the Q factor of the circuit.

There are two ways to perform the experiment: you could keep the frequency of the RF field constant and vary the magnetic field or you could keep the magnetic field constant and vary the frequency of the RF field. It is usually the former which is performed. A crucial factor is to have a highly homogeneous magnet to produce the constant field B_0 . If this is not the case, different parts of the sample will sit in slightly different magnetic fields and will come into resonance at slightly different points of the magnetic field sweep, causing the measured resonance to be extremely broad, maybe so much that it is washed out altogether.

Example 3.2

For a proton, $I = \frac{1}{2}$ and m_I can take the values $\pm \frac{1}{2}$ only, as illustrated in Fig. 3.10(b). The two states of the system are separated by an energy $\Delta E = g_N\mu_N B$ which is tiny: for a proton in a typical laboratory magnetic field $B_0 \sim 1$ T this corresponds to an energy splitting of $\sim 10^{-7}$ eV; when this energy is expressed as $k_B T$ it is equivalent to a temperature of only ~ 1 mK.

Therefore at room temperature, at this magnetic field, the nuclei will show only a minute tendency to line up with the applied magnetic field on average because the thermal randomizing energy will vastly dominate over the weak alignment energy. Any effect due to the magnetism of the nuclei would be practically impossible to detect if a resonant technique were not employed. Because in NMR one is perturbing the system with a RF signal of *exactly*⁵ the right frequency ω given by

$$\hbar\omega = \Delta E \quad (3.8)$$

we can excite transitions across this energy splitting. Why do we need radiofrequencies? The next example shows why.

Example 3.3

For a magnetic field of 1 T, the frequency ν is given by

$$\nu = \frac{\omega}{2\pi} = \frac{g_N \mu_N B_0}{2\pi \hbar} = 42.58 \text{ MHz} \quad (3.9)$$

which is in the radiofrequency region of the electromagnetic spectrum.

This means one can use oscillators and coils (in comparison with electron spin resonance experiments which have to use Gunn diodes and microwave waveguides, see Section 3.2.2). Therefore at exactly the right frequency of RF excitation the system will absorb energy. This is the nuclear magnetic resonance and is a resonance between the frequency of the RF excitation and the energy separation of the nuclear levels. Modern NMR spectrometers use 12–15 T magnets, so the frequency of the RF is in the ~ 500 – 650 MHz range, still in the radio frequency region of the electromagnetic spectrum.

The NMR resonance frequency at a given magnetic field can be slightly shifted up and down from the value $\gamma B/2\pi$ depending on the chemical environment of the nucleus. These chemical shifts (typically a few parts per million) are due to the fact that the electrons orbiting the nucleus slightly shield the nucleus from the applied field. The amount by which a nucleus is shielded in a given chemical environment is well known, allowing a given molecule to be ‘fingerprinted’. NMR lines can split due to magnetic coupling to neighbouring nuclei. This is known as *spin–spin coupling* and occurs via an indirect contact hyperfine interaction which is mediated by the electrons. This also gives information about the environment of the nucleus.

In order to understand the transitions which we are inducing in a little more detail, let us consider how a two-level spin system ($I = \frac{1}{2}$) will absorb energy from the RF coil. We will label the lower level $-$ and the upper level $+$. The probability of inducing a transition between the two levels is independent of whether the transition is from the lower to the higher level or *vice versa*. This is because the perturbing Hamiltonian $\hat{\mathcal{H}} \propto \hat{I}_x$, appropriate for an oscillating RF field, is Hermitian, so that

$$|\langle \psi_- | \hat{\mathcal{H}} | \psi_+ \rangle|^2 = |\langle \psi_+ | \hat{\mathcal{H}} | \psi_- \rangle|^2. \quad (3.10)$$

Hence the probability per unit time of these so-called *stimulated transitions* between levels $+$ and $-$ is independent of the direction of the transition and occurs at a rate W which is proportional to the size of the RF power used to excite transitions. At time t , if there are $N_-(t)$ spins in the lower level, $W N_-(t)$ will be excited per unit time into the upper level. If there are $N_+(t)$ spins in the upper level, $W N_+(t)$ will be excited per unit time into the lower level. Thus

$$\frac{dN_+(t)}{dt} = W N_-(t) - W N_+(t) \quad (3.11)$$

$$\frac{dN_-(t)}{dt} = W N_+(t) - W N_-(t) \quad (3.12)$$

which when subtracted gives

$$\frac{d}{dt}(N_+(t) - N_-(t)) = -2W(N_+(t) - N_-(t)), \quad (3.13)$$

which can be solved to give

$$N_+(t) - N_-(t) = (N_+(0) - N_-(0))e^{-2Wt} \quad (3.14)$$

An initial difference in population tends exponentially to zero when driven by a stimulated electromagnetic transition. Let us now define $n(t)$ by

$$n(t) = N_+(t) - N_-(t), \quad (3.15)$$

so that eqn 3.14 becomes

$$n(t) = n(0)e^{-2Wt} \quad (3.16)$$

The rate of absorption or emission of electromagnetic energy by the spin system is easily calculated since each transition involves an exchange of energy equal to $\hbar\omega$. The energy of the system at time t is

$$E(t) = N_-E_- + N_+(E_- + \hbar\omega), \quad (3.17)$$

where E_- is the energy of the lower level. This expression for $E(t)$ can be rearranged to give a constant term plus $\frac{1}{2}\hbar\omega n(t)$. Thus the rate of absorption of energy is given by

$$\frac{dE}{dt} = -W\hbar\omega n(t) \quad (3.18)$$

and, because it is proportional to $n(t)$, will tend to zero with a time constant of $1/2W$ as the populations of the upper and lower levels become progressively equalized. This demonstrates also that to absorb energy you need $n(t) \neq 0$, i.e. a population difference. This is produced by interaction with the thermal modes of the system, so that the polarization of the system is brought back towards a Boltzmann probability expression:

$$\left(\frac{N_+}{N_-}\right)_0 = e^{-\hbar\omega/k_B T} \quad (3.19)$$

Here, the subscript 0 denotes the equilibrium value of the system. After absorption of energy the polarization may change, but if the nuclear spin system interacts with the thermal motion and excitations of the sample, the polarization of the spin system can return towards its equilibrium value. If the polarization of the spin system has been reduced to zero by rapid electromagnetic transitions, the spin system will take some time to 'recover' once the electromagnetic transitions are switched off; this time is called T_1 and measures the time constant of the interaction between the spin system and its surroundings. T_1 is the **spin-lattice relaxation time** (sometimes called the longitudinal relaxation time). Thus we would expect that the polarization would be restored as

$$n(t) = n_0(1 - e^{-t/T_1}). \quad (3.20)$$

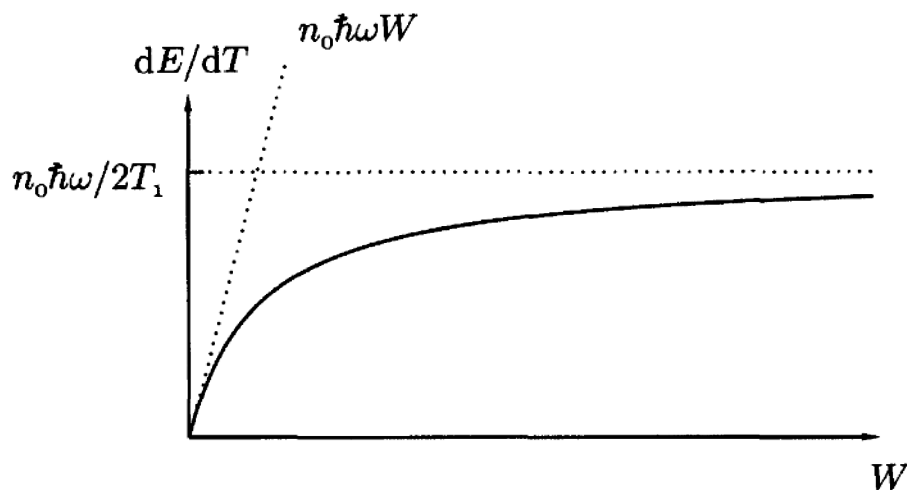


Fig. 3.11 The rate of absorption of electromagnetic energy given by eqn 3.25 as a function of the transition rate W . For low W , the expression is proportional to W , but at high W it saturates at a value governed by $1/T_1$.

Now let us put both processes in together, the stimulated transitions and the relaxation processes. Thus we may combine our equations to arrive at

$$\frac{d}{dt}n(t) = -2Wn(t) + \frac{n_0 - n(t)}{T_1}, \quad (3.21)$$

which has a steady state solution when

$$\frac{d}{dt}n(t) = 0. \quad (3.22)$$

Thus

$$n(t) = \frac{n_0}{1 + 2WT_1}, \quad (3.23)$$

and the steady state rate of absorption of electromagnetic energy is given by

$$\frac{dE}{dt} = n(t)\hbar\omega W \quad (3.24)$$

$$= n_0\hbar\omega \frac{W}{1 + 2WT_1}. \quad (3.25)$$

This relation is plotted in Fig. 3.11. For low-amplitude RF perturbing fields, the rate of absorption of electromagnetic energy is proportional to W . But at large RF perturbing fields, it settles at a level proportional to T_1 , and independent of the precise magnitude of W . This is known as **saturation**.

The rate of absorption of electromagnetic energy is proportional to the population difference between the upper and lower levels. The rate of absorption may therefore be used to monitor the population difference and to observe its time dependence, subject of course to the constraint that such observation will necessarily perturb the population difference to some extent. Equation 3.25 and Fig. 3.11 demonstrate that you can only increase the power of the RF signal so much to improve the signal. This also demonstrates that dE/dt is proportional to $\hbar\omega$; more energetic (i.e. higher frequency) photons lead to a larger signal. But of course this means larger magnetic fields to match $g\mu_N B$ to $\hbar\omega$. This will prove to be an advantage for ESR (see Section 3.2.2 below) since very beefy microwave photons are needed to excite transitions across the larger electronic

gaps, and this leads to a higher sensitivity. Because electronic moments are larger than nuclear moments, you can get away with smaller magnetic fields than NMR and still use higher frequency photons.

Let us return to NMR and consider relaxation processes in a little more detail. In a magnetic field a very weak polarization of the spins will exist (weak because the energy separation $\Delta E \ll k_B T$.) Let us say it takes an equilibrium value M_0 with the static magnetic field $\mathbf{B}_0$ switched on (recall that $\mathbf{B}_0$ is along the z axis). The effect of the radio frequency excitations will be to progressively destroy this magnetization. Also, as we have seen, if the magnetic field is switched off, the magnetization will relax back to the equilibrium value by the weak interactions with the surroundings with a time constant T_1 . Thus we expect that

$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1}. \quad (3.26)$$

This T_1 relaxation must involve interactions with the lattice because energy must be exchanged with it. This is because changing M_z has energetic consequences since relaxing a spin in the z -direction changes its orientation with respect to the applied field, and hence changes its energy.

The M_x and M_y components should be zero, but if they are not they will relax back to zero in a time T_2 such that

$$\frac{dM_x}{dt} = -\frac{M_x}{T_2}, \quad \frac{dM_y}{dt} = -\frac{M_y}{T_2}. \quad (3.27)$$

T_2 is the **spin-spin relaxation time** and is the characteristic time for dephasing because it corresponds to the interaction between different parts of the spin system. It can also be due to inhomogeneities in the magnetic field B_0 . It thus leads to differences in precession frequency due to the interactions between the observed spin and the spins of its neighbours. Changing M_x or M_y has no energetic consequences because the applied field is along M_z .

The effect of the applied magnetic field is also to cause spin precession, so that the equations for M_x , M_y and M_z are

$$\frac{dM_x}{dt} = \gamma(\mathbf{M} \times \mathbf{B})_y - \frac{M_x}{T_1}, \quad (3.28)$$

$$\frac{dM_y}{dt} = \gamma(\mathbf{M} \times \mathbf{B})_x - \frac{M_y}{T_1}, \quad (3.29)$$

$$\frac{dM_z}{dt} = \gamma(\mathbf{M} \times \mathbf{B})_z + \frac{M_0 - M_z}{T_1}, \quad (3.30)$$

Felix Bloch (1905–1983)

which are known as the **Bloch equations**. They were worked out in 1946 and can be solved for a number of cases of interest, very often by using a rotating reference frame method in which the coordinates are changed to ones which rotate in the xy plane at the resonance frequency.

The spin-spin relaxation time T_2 can be measured using the following technique: in thermal equilibrium with B_1 switched off, there is a weak magnetization parallel to z . A short pulse of RF signal of duration t_p , with the frequency of the RF signal set close to the resonance frequency of the spins, causes the spins to rotate by an angle $\gamma B_1 t_p$ where B_1 is the amplitude of the RF signal. This angle can be adjusted by changing t_p and if $t_p = \pi/2\gamma B_1$, it produces what is known as a **90° pulse**. The amplitude and time of the pulse are chosen



Fig. 3.12 A simulated free induction decay signal. The oscillations die away with time due to spin-spin relaxation (T_2 processes). The data show an interference pattern which results from three inequivalent nuclei, each of which experiences a slightly different magnetic field because of their different sites within a crystal.

so that the magnetization just precesses for a short period and rotates into the xy plane. With B_1 switched off, the magnetization now precesses in the static field B_0 , rotating steadily in the xy plane at a frequency γB_0 , producing an induced voltage in the coil. The oscillation would carry on for ever if it were not for T_2 relaxation processes (spin-spin relaxation), which cause the oscillations to relax, thereby giving a measurement of T_2 . The corresponding decay of voltage in the coil is known as **free induction decay** (see Fig. 3.12).

One problem that occurs is that this method also measures relaxation due to the inhomogeneity of the magnet because this will cause different parts of the sample to give slightly different precession frequencies. What is needed is a method to separate the real spin-spin relaxation from the relaxation due to the inhomogeneity of the magnet. One way this can be achieved is to use a **spin echo technique**. The principle of this technique is illustrated in Fig. 3.13 and involves both a 90° pulse (which tips the spins by 90° into the xy plane as before) and the subsequent application of a 180° pulse (which rotates the spins by 180°). The effect can be likened to a marathon in which all athletes run steadily but some athletes are faster than others. After the starter's pistol (the 90° pulse) the athletes begin to run as a pack, but after a while the fitter begin to lead while others begin to trail behind. Then suddenly, at time τ , the rules of the game are changed (the 180° pulse) and the runners are told to turn around and head for the start of the race. Now the slower runners have an advantage since they are closer to the start. But by symmetry it is easy to see that the runners will all arrive at the start of the race simultaneously, at time 2τ , regardless of their individual speeds. In the same way, all the spins realign after 2τ , regardless of their individual dephasing. This is the spin echo.

Thus in the experiment we can use the spin echo to remove the problem that the magnetic field is inhomogeneous causing spins to precess at different rates. Also the spread in fields due to chemical shifts is removed. But we will not have taken away the effect of the spin-spin interaction which is due to time-dependent fluctuating random magnetic fields due to neighbouring nuclei. This relaxation mechanism cannot be refocussed and so the echo signal will be reduced in amplitude by an amount which depends on τ . The NMR intensity of the echo signal should follow $I(2\tau) = I(0)e^{-2\tau/T_2}$ and so a measurement of the true T_2 , due to the spin-spin coupling, is possible.

A measurement of T_1 can be performed using a similar method, though this time a 180° pulse is applied first. This causes the magnetization to rotate from

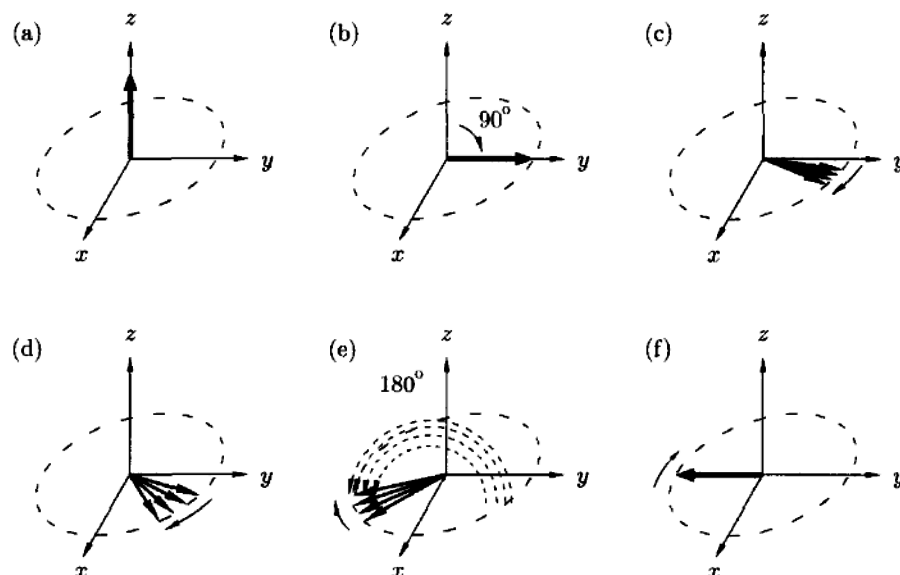


Fig. 3.13 The spin echo effect. (a) The equilibrium magnetization initially lies along the z direction, parallel to B_0 . (b) At a time which we will call $t = 0$, a 90° pulse along the x axis is used to rotate the spins into the xy plane. (c) Because of the steady field B_0 along the z axis, the spins now precess in the xy plane, though at slightly different rates because of inhomogeneities in the magnetic field. (d) This means they progressively dephase with respect to each other, becoming more separated in angle as they precess round. (e) At time $t = \tau$, a 180° pulse along the x axis rotates the spins by 180° around the x axis, and as they subsequently precess in the field B_0 , their order has been reversed. (f) They then will come back together at time 2τ , producing a spin echo signal, provided no other relaxation process has occurred in the meantime. Although the figure has been drawn for the case of a short delay between the 90° pulse and the 180° , the effect will work if the delay is long and the spins make a number of rotations before the 180° pulse is applied.

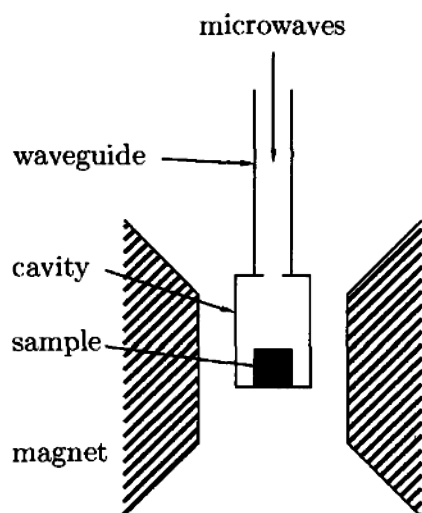


Fig. 3.14 Schematic diagram of an ESR experiment. Microwaves enter a cavity via a waveguide and the absorption of microwaves induced by a resonance is measured by monitoring the Q -factor of the cavity. The sample must be placed in the centre of the magnet, where the field is most clearly uniform.

along $\hat{z}$ to along $-\hat{z}$. It then relaxes back to $\hat{z}$ with a time constant T_1 but does not precess. Thus as a function of time τ after the 180° pulse

$$M(\tau) = \hat{z}M_z(\tau) = \hat{z}M_0(1 - 2e^{-\tau/T_1}). \quad (3.31)$$

At a time τ after the 180° pulse, a 90° pulse is used to rotate the magnetization into the xy plane where it now begins to do a free induction decay with initial amplitude $M_z(\tau)$. Thus by measuring the initial amplitude of the free induction decay as a function of the time delay between the 180° pulse and the 90° pulse, T_1 can be deduced. A variety of other pulse sequences can also be used to extract additional types of information.

3.2.2 Electron spin resonance

One can perform the analogous experiment to NMR with electrons. In this case the effect is known as **electron spin resonance (ESR)**, or sometimes as **electron paramagnetic resonance (EPR)**. The electron magnetic moment is much larger than that of nuclei so the precession frequencies are also much higher. For typical laboratory magnetic fields, the electromagnetic radiation is now in the microwave regime. A schematic of the experimental setup is shown in Fig. 3.14. The sample sits in a resonant cavity and microwave radiation enters via a waveguide. It is usually much more convenient to keep the frequency of

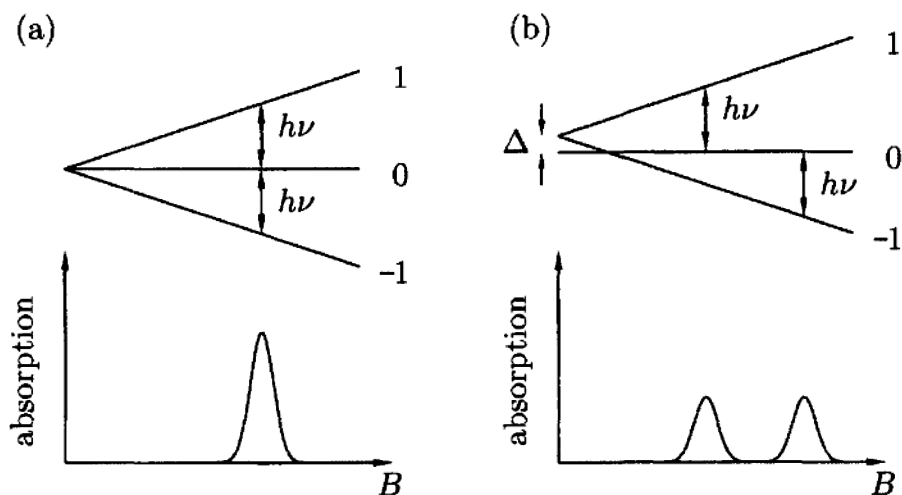


Fig. 3.15 ESR in an ion with $J = 1$ such as Ni^{2+} with (a) no crystal field splitting (leading to a single ESR line) and (b) crystal field splitting Δ (leading to two ESR lines).

the microwave radiation fixed and to sweep the magnetic field. The cavity has a very high quality factor (Q -factor) and is designed to enhance the sensitivity of the detection of the weak ESR signal. The microwave absorption is modified as a function of the applied magnetic field. Very often a set of modulation coils is added to provide a small oscillatory magnetic field at the sample in addition to the static magnetic field provided by the magnet. The measurement can then be further enhanced by a phase sensitive detection technique.⁶

In an ESR transition the selection rule is

$$\Delta m_J = \pm 1, \quad (3.32)$$

the familiar dipole selection rule because one is inducing magnetic dipole transitions. Figure 3.15 shows these transitions in an ion with $J = 1$ such as Ni^{2+} in an octahedral environment. The experiment is performed at fixed frequency $h\nu$ with the magnetic field swept. Every time two adjacent levels are separated by $h\nu$, there is an ESR transition. Figure 3.15 demonstrates how the position and the number of lines can give information about the crystal field splitting.

Figure 3.16 shows the energy of a free Mn^{2+} ion (spin- $\frac{5}{2}$) as a function of magnetic field. At a magnetic field B this gives a number of possible transitions between the different m_J levels, all of which have size $g\mu_B B$. Thus a single line might be expected to be observed in an ESR experiment at a frequency ν given by $h\nu = g\mu_B B$. However, in a real experiment on a solid containing Mn^{2+} ions, the crystal field significantly complicates this picture, even in a cubic environment.

However, the hyperfine coupling between the nucleus and the electron gives a term in the Hamiltonian equal to $A\mathbf{I} \cdot \mathbf{J}$ and each m_J level is split into $2I + 1$ hyperfine levels depending on the value of m_I . In an ESR transition the selection rules are therefore given by eqn 3.32 and by

$$\Delta m_I = 0. \quad (3.33)$$

⁶A lock-in amplifier is used to produce a signal which drives the oscillatory magnetic field. The microwave signal is then fed into the lock-in amplifier, which rejects any component of this signal which is not at the frequency of the oscillatory magnetic field. In this way the noise is drastically reduced.

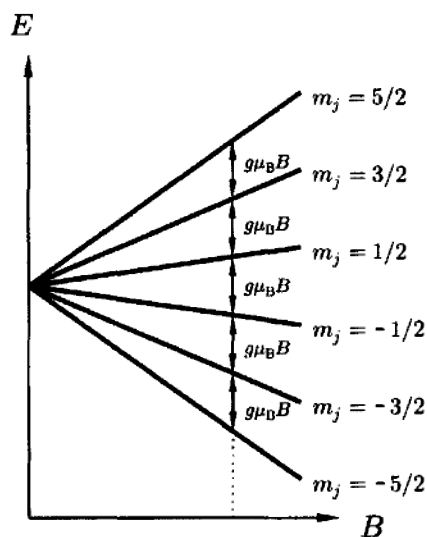


Fig. 3.16 The energy of a spin- $\frac{5}{2}$ magnetic moment, such as a free Mn^{2+} ion, as a function of magnetic field. The possible transitions with the selection rule $\Delta m_J = \pm 1$ are as shown. This picture ignores the crystal field.

⁷This is because the energy spacing between the nuclear levels is very small in comparison with the energy of the microwave photons.

The latter selection rule occurs because the microwave frequency induces only dipole transitions between electronic levels, not between nuclear levels.⁷ Since A is very small in comparison with $g\mu_B B$ for typical fields in an ESR experiment, one need only compute the components of $\mathbf{J}$ parallel to $\mathbf{B}$ so that the energy levels in a magnetic field are

$$E = g\mu_B B m_J + A m_I m_J \quad (3.34)$$

where the energy of the nuclear spin in the applied magnetic field has been ignored (since it is much smaller even than the hyperfine energy). Using the selection rule, one can now show that the resonances should occur at

$$\begin{aligned} h\nu &= [g\mu_B B(m_J + 1) + A m_I(m_J + 1)] - [g\mu_B B m_J + A m_I m_J] \\ &= g\mu_B B + A m_I \end{aligned} \quad (3.35)$$

and each ESR line will split up into $2I + 1$ hyperfine lines.

Example 3.4

For the case of a free Mn^{2+} ion, $I = \frac{5}{2}$, so this means six lines in the ESR spectrum. This situation is depicted in Fig. 3.17 (although for a Mn^{2+} ion in a crystal, the crystal field causes the splittings to be more complex and to depend on the angle between the magnetic field and the crystal axes in a non-trivial way).

ESR experiments probe electronic spins in a useful range because crystal field splittings are very often at GHz frequencies. ESR studies in paramagnetic salts have provided a great deal of detailed information concerning the crystal and ligand fields. Very often the effect is anisotropic so that lines move as the magnetic field is rotated with respect to the crystal axis.

Example 3.5

As an example of how this information can be obtained, consider Ce^{3+} ($4f^1$) in an environment with axial symmetry (see Fig. 3.18). This ion has $L = 3$ and $S = \frac{1}{2}$ so the spin-orbit interaction splits this into an eightfold degenerate $J = \frac{7}{2}$ level and a lower sixfold degenerate $J = \frac{5}{2}$. The crystal field is axial and contains only even powers of J_z by symmetry, so the sixfold $J = \frac{5}{2}$ manifold (as a group of lines are called) splits up into three doublets, and the eightfold $J = \frac{7}{2}$ manifold into four. These splittings are much smaller than the spin-orbit interaction in rare earth systems. The doublets can be split by applying a magnetic field which is almost always the smallest energy term for typical laboratory magnetic fields, usually around 10^{-4} eV. At low temperature, only the ground state will be occupied and it is this transition which will be observed.

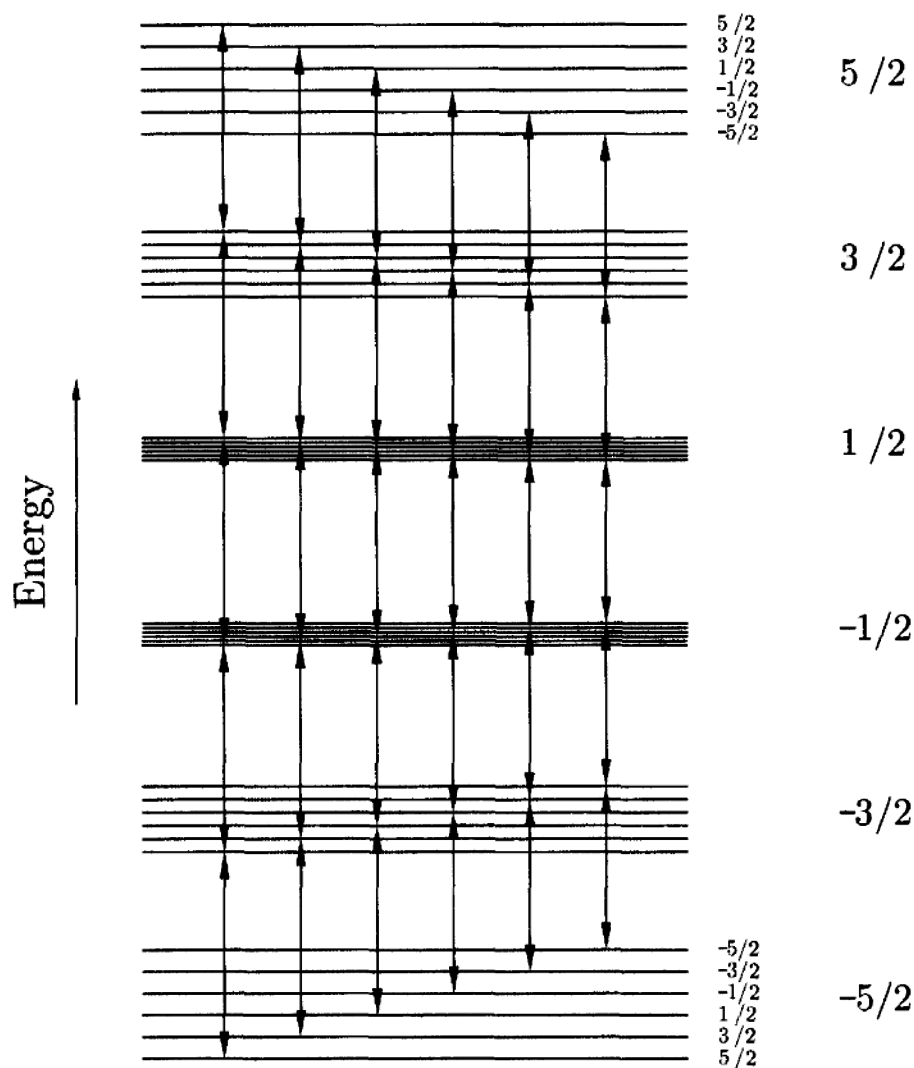


Fig. 3.17 The hyperfine splitting in a free Mn^{2+} ion at a fixed magnetic field. $J = \frac{5}{2}$ and $I = \frac{5}{2}$ so each of the six m_J levels is split into six hyperfine levels. The possible transitions with $\Delta m_J = \pm 1$ and $\Delta m_I = 0$ are shown, giving rise to six distinct lines in the ESR spectrum. This picture ignores the effect of the crystal field which leads to these splittings becoming strongly dependent on the direction of the applied magnetic field with respect to the crystalline axes.

The ground state of many systems is, like Ce^{3+} , a doublet. This is a consequence of **Kramers theorem**, which states that in a system containing an odd number of electrons, at least two-fold degeneracy must remain in the absence of a magnetic field. The pairs of states involved, **Kramers doublets**, are time conjugate⁸ and therefore can be split by a magnetic field but not by an electric field. Ce^{3+} ($4f^1$) has an odd number of electrons in the $4f$ shells and therefore qualifies as a Kramers ion.

The ground state of Ce^{3+} is a Kramers doublet and, viewed in isolation, looks like a spin- $\frac{1}{2}$ state, although its quantum numbers are very different. It can in fact be assigned an **effective spin** $\tilde{S} = \frac{1}{2}$. This is a fictitious angular momentum which is chosen so that the degeneracy of the group of levels under consideration is set equal to $(2\tilde{S} + 1)$. The motivation for this substitution is to find an effective spin Hamiltonian that approximately describes the experimental situation in the language of something much simpler, namely a free atom or ion with spin- $\frac{1}{2}$. This Hamiltonian can be written

$$\tilde{H} = g\mu_B \mathbf{B} \cdot \tilde{\mathbf{S}} \quad (3.36)$$

Hendrik A. Kramers (1894–1952)

⁸that is, they are complex conjugates of each other, and are thus time reversed versions of each other.

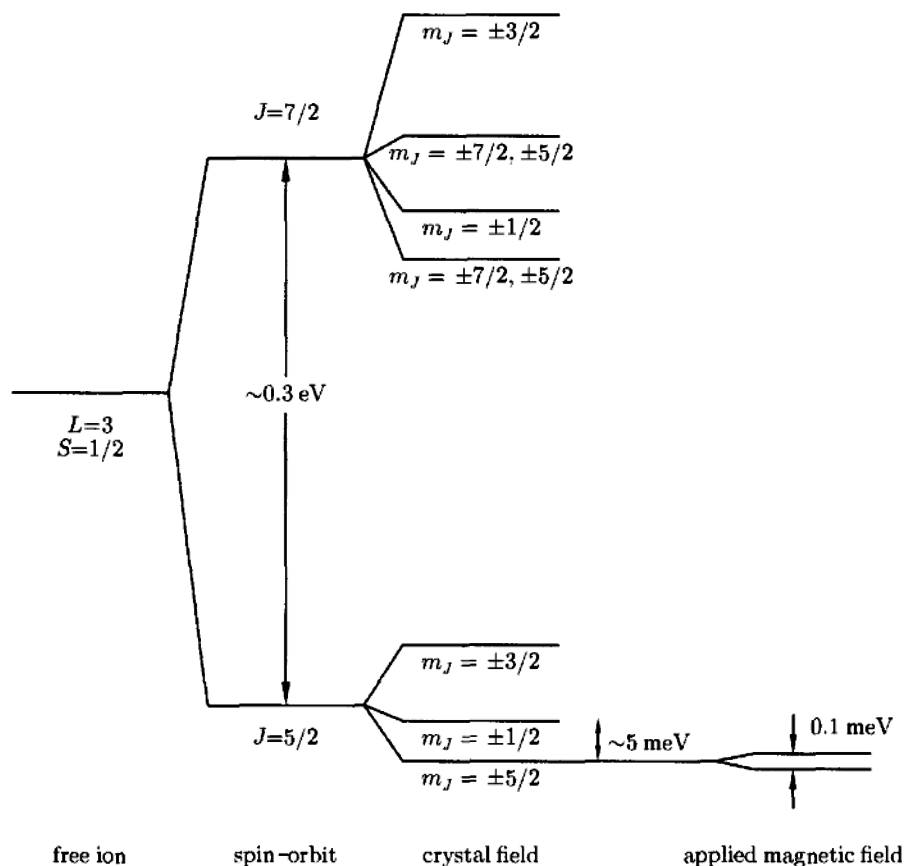


Fig. 3.18 The energy levels for a Ce^{3+} ion. The dominant energy splitting is the spin-orbit interaction (this energy corresponds to about 3000 K so the upper manifold is not appreciably thermally excited). The two manifolds are split into sets of doublets by an assumed axial crystal field. The crystal field splittings have been exaggerated in this diagram so that they are easily observable. The lower splitting is $\sim 5 \text{ meV}$ which corresponds to about 50 K. Finally an applied magnetic field splits all the doublets (only the lower one is shown split here).

where g is now an effective g -factor (sometimes called a spectroscopic splitting factor) which would be equal to 2 for a real isolated spin- $\frac{1}{2}$ electron but includes the orbital contribution in this case. Very often one needs to use an effective g -tensor, $\mathbf{g}$, to express the fact that the interaction of the spin with the field depends on the orientation of the magnetic field, so that eqn 3.36 must be replaced with

$$\tilde{H} = \mu_B (\mathbf{B} \cdot \mathbf{g} \cdot \tilde{\mathbf{S}}) = \sum_{ij} g_{ij} B_i \tilde{S}_j. \quad (3.37)$$

The effect of the crystal field can then be included by adding in a term to the Hamiltonian to represent the energetic preference for the spin to lie along particular crystalline directions because of the crystal field. This is known as single ion anisotropy and the term in the Hamiltonian, $\tilde{H}_{\text{SI}}$, is

$$\tilde{H}_{\text{SI}} = -D S_z^2, \quad (3.38)$$

for a uniaxial crystal (i.e. one with a particular axis such that the energy just depends on the angle of the spin with that axis) and for a cubic crystal

$$\tilde{H}_{\text{SI}} = -D(S_x^4 + S_y^4 + S_z^4) \quad (3.39)$$

where D is an anisotropy constant.

ESR is useful not only in studying the physics of salts with paramagnetic ions but is used extensively in chemistry. The resonances can also be extremely

sensitive to atomic position in a molecule and the technique is greatly used in the study of chemical reactions and particularly of free radicals, which are atoms or fragments of molecules with an unpaired electron and are often highly reactive. Free radicals are of great environmental and biological importance. ESR is not as widely used as NMR because it is applicable only to materials with an unpaired spin.

Example 3.6

One example of a field in which ESR is very useful is in studies of various biological molecules which often contain transition metal ions. An important example is haem, an iron porphyrin group, which is found in haemoglobin and myoglobin. These molecules are involved in transfer of oxygen and other small molecules such as carbon monoxide. The Fe(II) ion in the haem converts from $S = 2$ to $S = 0$ after binding to the small molecule (this is an example of a spin transition, see Section 3.1.1).

The spin-lattice relaxation times in ESR are often very short in comparison with those of NMR; this is because the electronic moment is much more strongly coupled to lattice vibrations than are the nuclei. Lattice vibrations cause a modulation in the crystal field which couples to the electronic moment via the spin-orbit interaction, often making T_1 so short that the resulting ESR line is too broad to observe. (The width of the line scales as T_1^{-1}). A favourable case is Mn^{2+} which is an S-state ion ($3d^5$, $^6S_{5/2}$) and therefore has no orbital moment to which the spins can couple and ESR can be observed easily in this ion at room temperature. For other transition metal ions, the size of T_1 depends on the degree of orbital quenching.⁹ The resonance can be narrowed (T_1 increased) by cooling the sample, since this reduces the lattice vibrations. This also has the benefit of increasing the relative thermal population of the levels which increases the intensity of the resonance.

A variant of ESR is a technique called ENDOR (an acronym for Electron Nuclear DOuble Resonance) which can be used to measure hyperfine interaction constants and the nuclear Zeeman interaction with great precision. It is often impossible to measure directly NMR transitions between nuclear levels in paramagnetic salts because they are very weak. In ENDOR one uses both an RF signal and a microwave signal and capitalizes on the high sensitivity of ESR. The microwave power is tuned to a $\Delta m_I = 0$ transition and the ESR signal then depends on the relative population of those levels. The populations can be changed by using RF power tuned to the NMR transition. The NMR transition can then be detected via its effect on the ESR signal.

⁹The more completely the orbital moment is quenched towards zero, the smaller the coupling of the electronic moment to the lattice vibrations and this leads to a longer T_1 and hence a narrower ESR line.

3.2.3 Mössbauer spectroscopy

Another form of spectroscopy is based upon the Mössbauer effect which was discovered in 1957. The principle of the experiment is illustrated in Fig. 3.19. A source containing ^{57}Co nuclei provides a ready supply of excited ^{57}Fe nuclei; these decay to the ground state via a gamma ray cascade which includes a

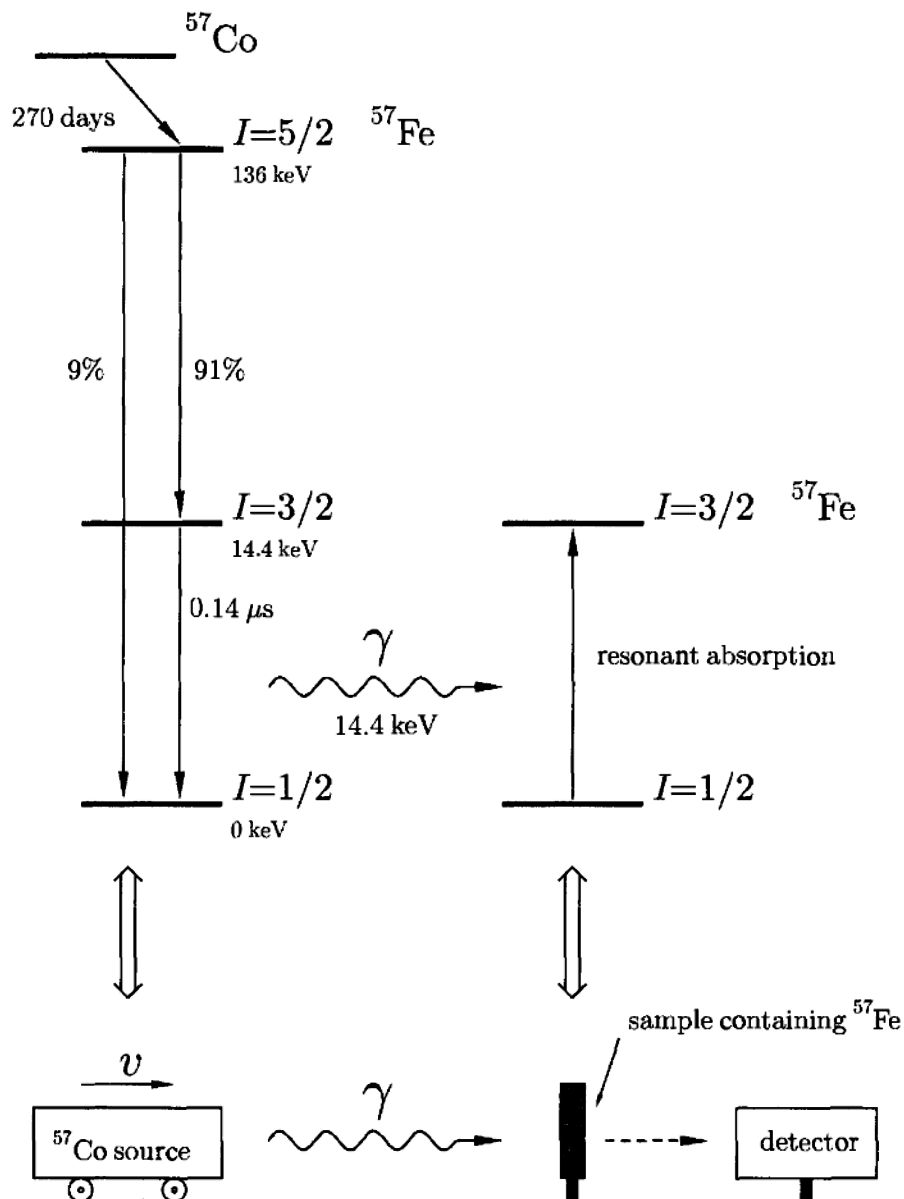


Fig. 3.19 Schematic illustration of the principles of the Mössbauer technique. ^{57}Co decays slowly into an excited state of the ^{57}Fe nucleus. The majority (91%) of the subsequent decay is rapidly to the $I = \frac{3}{2}$ state which then decays to the $I = \frac{1}{2}$ ground state. This decay releases a 14.4 keV photon. The experiment is performed by moving the ^{57}Co source at a speed v relative to the sample which must contain some ^{57}Fe nuclei. The detector measures the transmission of γ -rays through the sample which can be used to deduce the absorption.

14.4 keV gamma ray (corresponding to a frequency $\nu = 3.5 \times 10^{18} \text{ Hz}$). This gamma ray can excite a transition in the sample being studied if it is absorbed resonantly. To do this, its energy must match the energy gap in the sample. By moving the source at speed v one can very slightly adjust the frequency of the gamma ray because of the Doppler effect. Because of the high frequency of the photon, the Doppler shifts can be quite significant: a velocity $v = 1 \text{ mm s}^{-1}$ leads to a shift of $\nu v/c \approx 12 \text{ MHz}$. Thus one can probe any splittings in the ground state in the source or absorber nucleus which might result from magnetic or other interactions.

The technique would be useless if the γ ray photon emitted by the excited ^{57}Fe nucleus did not have a well defined frequency. This is why the relatively

slow decay of the ^{57}Fe nucleus is vital: the rather plodding $0.2\ \mu\text{s}$ half-life corresponds to an uncertainty of only about 2 MHz in the frequency. A second vital feature is that the ^{57}Fe atoms are in the solid state. In order to conserve momentum, a free Fe atom would be subject to a recoil velocity of the order of $h\nu/m_{\text{Fe}}c \sim 80\ \text{m s}^{-1}$ which would ruin the experiment (we have seen that even a relative velocity of $1\ \text{mm s}^{-1}$ can produce a measurable effect). However a ^{57}Fe atom which is held rigidly in a solid transmits that momentum to the entire crystal. If the recoil energy is lower than a certain quantity, the probability of phonon emission becomes vanishingly small and the gamma ray can be emitted without any loss of recoil energy. This recoil-free emission and resonant absorption of γ -rays is the essence of the **Mössbauer effect**.

Rudolf L. Mössbauer (1929–)

The conditions described above therefore imply that the effect will be optimized for low-energy γ -rays associated with nuclei strongly bound in a crystal lattice at low temperatures. ^{57}Fe is not the only isotope which is ideal for this purpose (other isotopes which can be used include ^{119}Sn , ^{127}I , ^{151}Eu , and ^{197}Au) but it is the most commonly used. The raw result which one obtains from a Mössbauer measurement is a plot of γ -ray counts (or relative absorption) against the velocity of the source with respect to the absorber.

Why should this effect tell us anything about the sample? For a start, the resonant absorption may not occur exactly where you would expect it to (i.e. when the source is stationary), but may be slightly shifted (i.e. to when the source is moving at a particular velocity). This **isomer shift** is due to the slight change in the Coulomb interaction between the nuclear and electronic charge distributions over the nuclear volume which is associated with the slight increase of size of the ^{57}Fe nucleus in the $I = \frac{3}{2}$ state.

Furthermore, one may not necessarily just observe one resonant absorption line as a function of source velocity, but perhaps a number of lines. This can be due to **quadrupole splitting** or **magnetic splitting**. The first effect is due to the electric quadrupole moment of the excited ^{57}Fe nucleus (although the ground state of ^{57}Fe has $I = \frac{1}{2}$ and thus has no electric quadrupole moment, the excited state of interest has $I = \frac{3}{2}$ and nuclei with $I > \frac{1}{2}$ can have a non-zero quadrupole moment). If the nucleus is subjected to an electric field gradient, as may be found in certain crystal environments, the interaction between the nuclear quadrupole moment and the electric field gradient splits the excited $I = \frac{3}{2}$ state into a doublet, so that two lines are produced in the Mössbauer spectrum. The second effect is caused by the interaction between the nucleus and the local magnetic field. This can split the $I = \frac{1}{2}$ ground state into a doublet, and the excited $I = \frac{3}{2}$ into a quadruplet, leading to six possible lines in the Mössbauer spectrum (the selection rule is $\Delta m_I = 0, \pm 1$). This effect can be used to detect magnetic exchange interactions and local magnetic fields. All these shifts and splittings are illustrated schematically in Fig. 3.20. Note that the figure is not (and could not be) drawn to scale. The observed splittings are typically in the range 10^7 – 10^8 Hz and are therefore 11 or 12 orders of magnitude smaller than the energy gap between the ground ($I = \frac{1}{2}$) and excited state ($I = \frac{3}{2}$) of the ^{57}Fe nucleus. Putting this another way, one is measuring energy splittings of less than $1\ \mu\text{eV}$ with a 14.4 keV photon! Were it not for the Mössbauer effect, the remarkable recoil-free resonance absorption and emission of these γ rays, such experiments would be completely impossible.

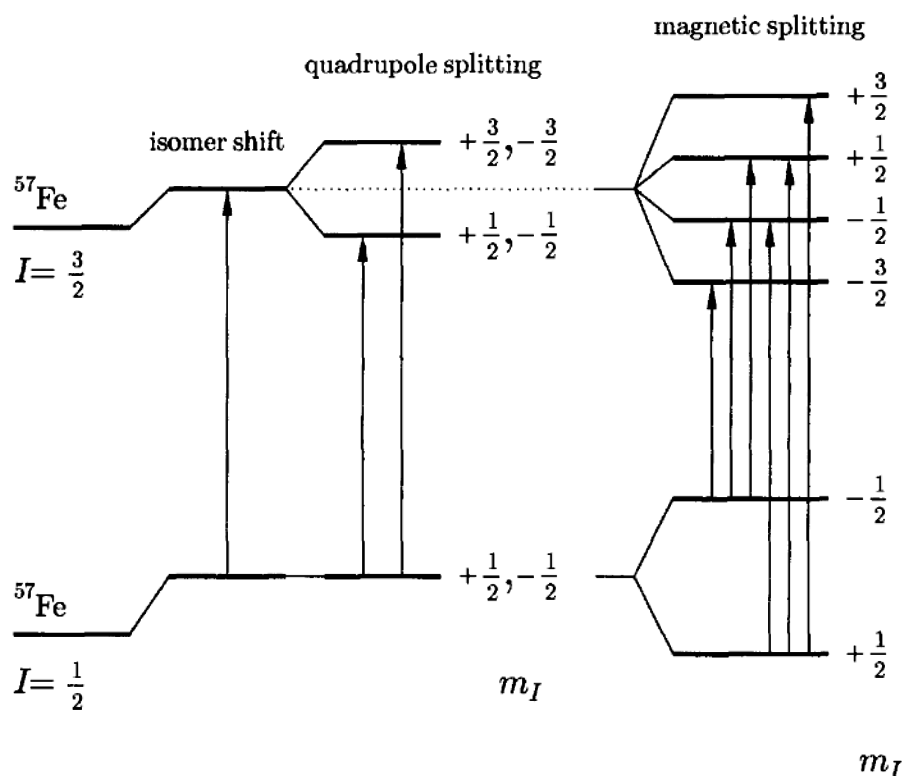


Fig. 3.20 The effects of chemical shift, quadrupole splitting and magnetic splitting on the nuclear energy levels of ^{57}Fe . The arrows show the Mössbauer absorption transitions. The difference in the size of the transitions is greatly exaggerated; in reality they typically differ from each other by less than 1 part in 10^{11} .

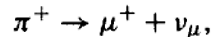
3.2.4 Muon-spin rotation

The muon is a spin- $\frac{1}{2}$ particle (charge $\pm e$, mass $250m_e$) which has a lifetime of $2.2 \mu\text{s}$. The muon is found in nature as the dominant constituent of cosmic rays arriving at sea-level. Muons can be used to study the magnetic properties of samples, but cosmic rays do not provide a sufficiently intense source. For this purpose it is therefore necessary to use the more intense beams of muons available from synchrotrons and cyclotrons.

A technique called muon-spin rotation (often abbreviated to μSR) will now be described in some detail. It is important to realize that in sharp contrast to the neutron and X-ray techniques that will be discussed later in the book, scattering is not involved; muons are implanted into a sample of interest and reside there for the rest of their short lives, never to emerge again. It is the positrons into which they decay that are released from the sample and yield information about the muons from which they came.

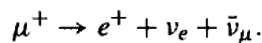
The mass of the muon is intermediate between that of the electron and the proton, and thus so are its magnetic moment and gyromagnetic ratio. The muon comes in either charge state, although it is the positive muon, μ^+ , which is of particular use in experiments in magnetism. As a small, positively charged particle, it is attracted by areas of large electron density and stops in interstitial sites in inorganic materials or bonds directly on to organic molecules. By contrast the negative muon, μ^- , implants close to an atomic nucleus and is generally much less sensitive to magnetic properties.

Muons can be prepared by colliding a high energy proton beam with a suitable target which produces pions. The pions (π^+) decay very quickly (26 ns) into muons



so that if one selects the muons arising from pions which have stopped in the target, the muon beam emerges completely spin-polarized. This is because the neutrino (ν_μ) which is also produced has its spin antiparallel to its momentum; the pion has no spin, so the muon must have its spin antiparallel to its momentum and hence emerges spin-polarized. These muons can then be implanted into a sample but their energy is large, at least 4 MeV. Following implantation they lose energy very quickly (in 0.1–1 ns) to a few keV by ionization of atoms and scattering with electrons. A muon then begins to undergo a series of successive electron capture and loss reactions which reduce the energy to a few hundred eV in about a picosecond. If muonium (see Fig. 3.21), a hydrogen-like state consisting of μ^+ and e^- , is ultimately formed then electron capture ultimately wins and the last few eV are lost by inelastic collisions between the muonium atom and the host atoms. All of these effects are very fast so that the muon (or muonium) is thermalized very rapidly. Moreover the effects are all Coulombic in origin and do not interact with the muon-spin so that the muon is thermalized in matter without appreciable depolarization. This is a crucial feature for muon-spin rotation experiments. One may be concerned that the muon may only measure a region of sample which has been subjected to radiation damage by the energetic incoming muon. This does not appear to be a problem since there is a threshold energy for vacancy production, which means that only the initial part of the muon path suffers much damage. Beyond this point of damage the muon still has sufficient energy to propagate through the sample a further distance, thought to be about 1 μm , leaving it well away from any induced vacancies and interstitials.

In a μSR experiment the muons are stopped in the specimen of interest and decay after a time t with probability proportional to e^{-t/τ_μ} where $\tau_\mu = 2.2 \mu\text{s}$ is the lifetime of the muon. The muon decay is a three-body process



The decay involves the weak interaction and thus has the unusual feature of not conserving parity. This phenomenon (which also lies behind the negative helicity of the muon neutrino) leads to a propensity for the emitted positron (e^+) to emerge predominantly along the direction that the muon-spin was aligned along when the muon decayed.

The angular distribution of emitted positrons is shown in Fig. 3.22 for the case of the most energetic emitted positrons. In fact positrons over a range of energies are emitted so that the net effect is something not quite as pronounced. In an experiment a magnetic field can be applied perpendicular to the initial muon-spin direction. This can cause the muon-spin to precess. By repeating the experiment for many muons, one can therefore follow the polarization of an ensemble of precessing muons with arbitrary accuracy, provided one is willing to take data for long enough. A schematic diagram of the experiment is shown in Fig. 3.23(a). Consider a muon, with its polarization aligned antiparallel to its momentum, which is implanted in a sample. (It is antiparallel because of

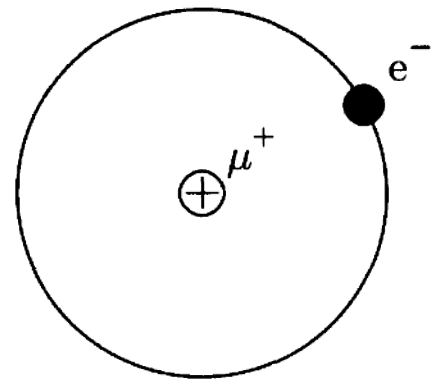


Fig. 3.21 Muonium, a hydrogen-like state consisting of μ^+ and e^- .

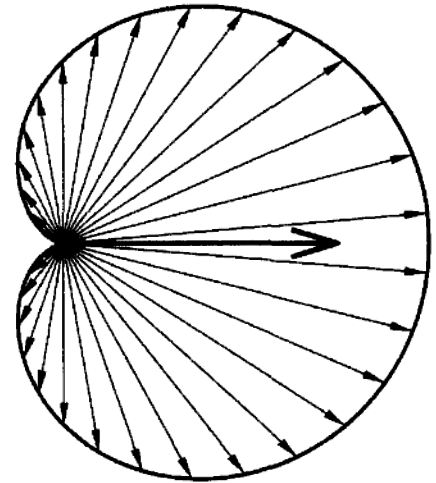


Fig. 3.22 The angular distribution of emitted positrons with respect to the initial muon-spin direction. The expected distribution for the most energetically emitted positrons is shown.

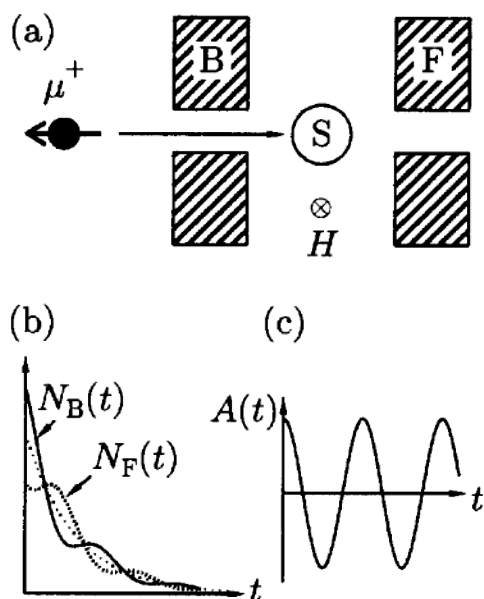


Fig. 3.23 (a) Schematic illustration of a μ SR experiment. A spin-polarized beam of muons is brought to rest in a sample S. Following decay, positrons are detected in either a forward detector F or a backward detector B. If a transverse magnetic field H is applied to the sample as shown then the muons will precess. (b) The number of positrons detected in the forward (broken-line) and backward (solid-line) detectors. The dotted line shows the average of the two signals. (c) The asymmetry function.

the way that it was formed, see above, so the muon enters the sample with its spin pointing along the direction from which it came.) If the muon is unlucky enough to decay immediately, then it will not have time to precess and a positron will be emitted preferentially into the backward detector. If it lives a little longer it will have time to precess so that if it lives for half a revolution the resultant positron will be preferentially emitted into the forward detector. Thus the positron beam from an ensemble of precessing muons can be likened to the beam of light from a lighthouse.

The time evolution of the numbers of positrons detected in the forward and backward detector are described by the functions $N_F(t)$ and $N_B(t)$ respectively and these are shown in Fig. 3.23(b). Because the muon decay is a radioactive process these two terms sum to an exponential decay. Thus the time evolution of the muon polarization can be obtained by examining the normalized difference of these two functions via the asymmetry function $A(t)$, given by

$$A(t) = \frac{N_B(t) - N_F(t)}{N_B(t) + N_F(t)}, \quad (3.40)$$

and is shown in Fig. 3.23(c). The muon is produced with 100% spin polarization so that, in contrast with NMR, one does not need to use tricks with pulses to observe a free induction decay. The muon will undergo Larmor precession in a magnetic field, internal or external. This precession can be followed by measuring the asymmetry. The frequency of precession is directly related to the magnetic field via $\omega = \gamma_\mu B$ where $\gamma_\mu = ge/2m_\mu = 2\pi \times 135.5 \text{ MHz T}^{-1}$ is the gyromagnetic ratio for the muon and m_μ is its mass (here $g \approx 2$). Implanted muons in magnetically ordered materials therefore precess in the internal magnetic field and directly yield oscillating signals whose frequency is proportional to the internal magnetic field. In this respect the muon behaves as a microscopic magnetometer. The Larmor precession frequencies for the proton (for NMR), electron (for ESR) and muon (for μ SR) are shown in Fig. 3.24.

The large magnetic moment of the muon makes it very sensitive to extremely small magnetic fields (down to $\sim 10^{-5} \text{ T}$) and thus is useful in studying

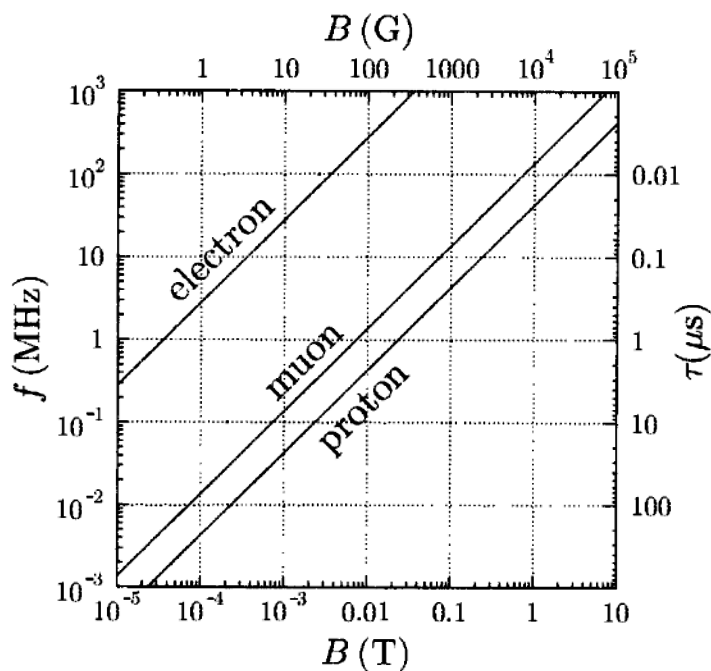


Fig. 3.24 The Larmor precession frequency f in MHz (and the corresponding period $\tau = 1/f$) for the electron, muon and proton as a function of applied magnetic field B .

small moment magnetism. It is also valuable in studying materials in which the magnetic order is random or of very short range. Since muons stop uniformly throughout a sample, each signal appears in the experimental spectrum with a strength proportional to its volume fraction, and thus the technique is helpful in cases where samples may be multiphase or incompletely ordered. Because no spatial information is obtained (in contrast to diffraction techniques considered later) single-crystal samples are not essential (though they can be useful in certain cases) and experiments can often provide information on the magnetic order of certain materials where conventional magnetic neutron diffraction cannot be simply performed. To extract quantitative information from μ SR experiments it is necessary to know the muon-site and this can in some cases hinder the search for a straightforward interpretation of the data. (In NMR of course the nucleus localizes the protons rather well so you exactly know the location of your probe.) Usually there is a small set of possible interstitial sites which the muon can occupy and in favourable circumstances only one will be consistent with the observed data. The technique has been widely applied to magnetic materials.

Further reading

- A good introduction to NMR may be found in P. J. Hore, *Nuclear magnetic resonance* OUP 1995. Also extremely useful is B. Cowan, *Nuclear magnetic resonance and relaxation* CUP 1997. The classic text on NMR is A. Abragam, *Principles of nuclear magnetism* OUP 1961.
- A. Abragam and B. Bleaney, *Electron Paramagnetic Resonance of Transition Ions*, Dover 1986, provides extensive information about crystal fields and ESR experiments in paramagnetic salts.
- The crystal field can be considered using the so-called Stevens operators, see K. W. H. Stevens, *Proc. Phys. Soc. A* **65**, 209 (1952) and M. T. Hutchings, *Solid State Physics* **16**, 227 (1966).
- For further information on the Mössbauer effect see *Mössbauer spectroscopy*, edited by D. P. E. Dickson and F. J. Berry, CUP 1986.
- For further information on μ SR see S. J. Blundell, *Contemp. Phys.* **40**, 175 (1999).

Exercises

- (3.1) A Sc^{++} ion has one electron in the 3d shell. It is in an anisotropic crystal and the crystal field can be written as a potential acting on the 3d electron as $A\hat{l}_z^2$. What are the lowest orbital states of the Sc ion if $A > 0$ and if $A < 0$? The spin-orbit coupling $\lambda\hat{l} \cdot \hat{s}$ is much smaller than the crystal field. When this is included, what are the approximate ground states of the ion, for $A < 0$ and $A > 0$? Discuss the effect on these states of applying a small magnetic field along the z axis and perpendicular to the z axis, and sketch the temperature dependence of the susceptibility.
- (3.2) Equal point positive charges are placed on each of the six corners of an octahedron. Taking the origin of a

set of cartesian coordinates to be at the centre of the octahedron, show that the potential close to the centre is given by

$$V = \frac{q}{4\pi\epsilon_0 a} \left[6 + \frac{35}{4a^4} \left(x^4 + y^4 + z^4 - \frac{3}{5}r^4 \right) + O\left(\frac{r^6}{a^6}\right) \right], \quad (3.41)$$

where q is the magnitude of each charge and a is the distance between the origin and each charge.

- (3.3) A compound has n transition-metal ions per unit volume, and each ion sits in a crystal field of the form

$$V = D \left(x^4 + y^4 + z^4 - \frac{3}{5}r^4 \right) \quad (3.42)$$

which acts as a perturbation on the degenerate d levels. We can choose the unperturbed wave functions in a variety of ways, so let us begin by picking eigenfunctions of $\hat{L}_z$, labelling them by their eigenvalue m_l . Thus

$$\begin{aligned} |\pm 2\rangle &= R(r) \sin^2 \theta e^{\pm 2i\phi} \\ |\pm 1\rangle &= \mp 2R(r) \sin \theta \cos \theta e^{\pm i\phi} \\ |0\rangle &= \sqrt{\frac{2}{3}} R(r) (3 \cos^2 \theta - 1), \end{aligned} \quad (3.43)$$

where $R(r)$ is the radial part of the wave function which includes a normalization constant. A general state of the system can be written as $|\psi\rangle = \sum_{m_l=-2}^2 a_{m_l} |m_l\rangle$ and then specified as a vector:

$$|\psi\rangle = \begin{pmatrix} a_2 \\ a_1 \\ a_0 \\ a_{-1} \\ a_{-2} \end{pmatrix}. \quad (3.44)$$

In this basis show that

$$V = \begin{pmatrix} A & 0 & 0 & 0 & 5A \\ 0 & -4A & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A & 0 \\ 5A & 0 & 0 & 0 & A \end{pmatrix} \quad (3.45)$$

where A is a constant and show that the eigenvalues and eigenfunctions are:

$$\begin{aligned} \text{eigenvalues:} & \quad 6A \quad 6A \quad -4A \quad -4A \quad -4A \\ \text{eigenfunctions:} & \quad |0\rangle \quad \frac{|2\rangle+|-2\rangle}{\sqrt{2}} \quad |1\rangle \quad |-1\rangle \quad \frac{|2\rangle-|-2\rangle}{\sqrt{2}} \end{aligned} \quad (3.46)$$

A magnetic field B induces a splitting $\mu_B B m_l$ so that V becomes

$$V = \begin{pmatrix} A + 2\mu_B B & 0 & 0 & 0 & 5A \\ 0 & -4A + \mu_B B & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A - \mu_B B & 0 \\ 5A & 0 & 0 & 0 & A - 2\mu_B B \end{pmatrix} \quad (3.47)$$

Calculate the eigenvalues and eigenvectors in this case. A plot of the eigenvalues is shown in Fig. 3.25.

Also show that at low temperatures, $k_B T \ll A$, the magnetic susceptibility in vanishing magnetic field is equal to

$$\chi = \begin{cases} \frac{2\mu_0 \mu_B^2 n}{3k_B T} & \text{for } A > 0 \\ \frac{2\mu_0 \mu_B^2 n}{5A} & \text{for } A < 0, \end{cases} \quad (3.48)$$

the first case being temperature independent and small, while the second case is equivalent to that expected for a free ion with $l = 1$.

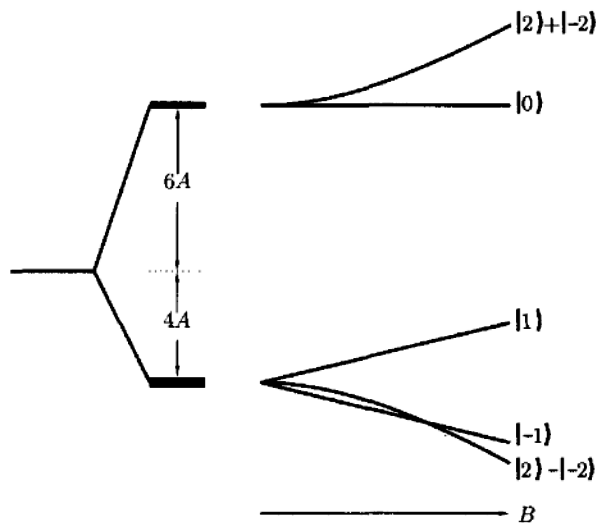


Fig. 3.25 A crystal field splits an ion with $l = 2$ into a triplet and an excited singlet. A magnetic field B splits these states further. The eigenstates as listed refer to vanishing magnetic field (as the field increases, the higher state acquires progressively more $|2\rangle$ character and the lower state acquires progressively more $|-2\rangle$ character).

- (3.4) An ion, whose nucleus has zero nuclear spin, has a ground state comprising two degenerate levels corresponding to an effective spin $S = 1/2$. The application of a magnetic field of flux density B produces a separation of the levels which is linear in B . A single electron paramagnetic resonance line is observed for the ion at a frequency of 30 GHz and a magnetic flux density of 0.6 T. An isotope with non-zero nuclear spin I gives rise to a hyperfine structure (described in the spin-Hamiltonian by a term $\mathbf{AI} \cdot \mathbf{S}$) comprising four approximately equally spaced resonance lines, with separation 10^{-2} T, symmetrically disposed about the line due to the isotope with zero nuclear spin. What information does this give about (a) the nuclear spin and (b) the nuclear magnetic moment of the isotope? Calculate the value of the parameter A .

Why is it sometimes necessary to perform these measurements at low temperature?

- (3.5) Figure 3.26 shows the magnetic moment per ion plotted versus B/T for three paramagnetic salts. Show that eqn 2.9 has the same form as the data at both low and high B/T . The 3d shells of free Cr^{3+} and Fe^{3+} ions contain three and five electrons respectively, and the 4f shell of Gd^{3+} contains seven. Using Hund's rules, calculate the values of S , L , J and g_J for these ions, and show that the values you obtain account well for the data in the figure for Gd^{3+} and Fe^{3+} , but not for Cr^{3+} . Infer the actual g-factor and ground-state quantum numbers for Cr^{3+} from the experimental data, and describe briefly

the reason for the discrepancy between the data and your initial calculation.

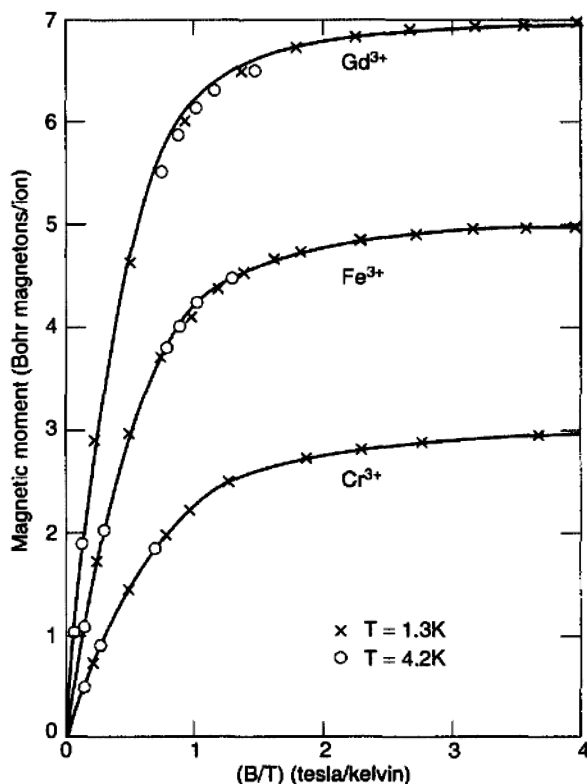


Fig. 3.26 The magnetic moment per ion for three paramagnetic salts: $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ (Cr^{3+}), $\text{NH}_4\text{Fe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ (Fe^{3+}) and $\text{Gd}_2(\text{SO}_4)_3 \cdot 8\text{H}_2\text{O}$ (Gd^{3+}). After W. Henry, *Phys. Rev.* **88**, 559 (1952).

- (3.6) An NMR spectrometer operates at a frequency of 60 MHz. At what applied magnetic field would you expect to observe the resonance of ^1H , ^2H , ^{13}C and ^{19}F nuclei? (Use the data in Table 2.3.)
- (3.7) An ESR spectrometer operates at a frequency of 9 GHz (known as X-band). What magnetic field would be required to observe a signal from the unpaired electron in DPPH (an organic molecule used to calibrate ESR spectrometers which gives a sharp signal at $g = 2$).
- (3.8) A spin Hamiltonian for a system is given by

$$\hat{\mathcal{H}} = g_{\parallel} \mu_B B_z S_z + g_{\perp} \mu_B (B_x S_x + B_y S_y) + D S_z^2. \quad (3.49)$$

Show that for two electrons in a triplet state ($S = 1$) in this system, the energy eigenstates E can be obtained from

$$E(E - D)^2 - E(g_{\parallel} \mu_B B \cos \theta)^2 - (E - D)(g_{\perp} \mu_B B \sin \theta)^2 = 0. \quad (3.50)$$

Find and sketch the solutions for $\theta = 0$ and $\theta = \pi/2$. Illustrate the possible photon transitions for $\hbar\omega > D$ and hence show how the parameters $g_{\parallel}$, $g_{\perp}$ and D could be obtained by measuring ESR for $\theta = 0$ and $\theta = \pi/2$.

- (3.9) In the Mössbauer effect a 14.4 keV photon is emitted from an ^{57}Fe atom. Calculate the velocity of recoil for a free ^{57}Fe atom and a ^{57}Fe atom that is rigidly held in the lattice of a 10 mg crystal. What is the Doppler shift of the 14.4 keV photon in each case? A sharp line in a Mössbauer experiment was obtained when the relative velocity between source and sample was 2 mm s^{-1} . Calculate the corresponding frequency shift in Hz and cm^{-1} and also the energy shift in eV.
- (3.10) The average muon lifetime is $2.2 \mu\text{s}$; assuming that a measurable fraction of muons live for $20 \mu\text{s}$, what lower limit does this put on the internal field that can be measured. In a typical experimental run, 10^7 muons are measured. Of these, how many live for $20 \mu\text{s}$ or longer? Pulses of muons are produced by a synchrotron with a pulse-width of 50 ns. What upper limit does this put on the internal magnetic field in a specimen that can be measured? (Hint: consider the precession signal resulting from the muons at the back and the front of the pulse.) At a cyclotron, a continuous beam of muons can be produced which avoids this problem. An ordered magnetic oxide has an internal field of 0.4 T at the interstitial site which is occupied by an implanted muon. What precession frequency do you expect to be measured in this case?
- (3.11) The saturation moment of an Fe^{3+} ion ($^6\text{S}_{5/2}$) in a crystal is expected to be $5\mu_B$ (see Fig. 3.26). But the effective moment deduced from the susceptibility measurements is expected to be $5.92\mu_B$ (see Table 3.1). Why the difference?